AI Data Centers: Public Ledger Stress-Test Toolkit

Technical companion to the public brief:

Who gets the asset, and who gets the bill?

For supporters

If Big Tech says it will pay its own way, show the contract that keeps households from paying later.

For opponents

If the contract is hidden, demand disclosure. The applicant should prove costs are not being shifted.

A project can be pro-growth and still need cost protections. A community can be pro-technology and still ask who pays.

Working release for public review and issue-spotting. Not a legal filing, expert report, or commission-ready evidentiary record.

Liquidity Happy Hours

Money View Lab research desk

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Toolkit Modules

Module	Use it for	Main tools
Module A. Screening the Project	First-pass review by residents, reporters, local officials, and public counsel.	Materiality triggers, project taxonomy, minimum viable review, proof levels, evidence ladder.
Module B. Electric and RTO Ledger	Utility, RTO, tariff, rate-base, and capacity-market exposure.	Ledger Adequacy Test, haircuts, stress scenarios, PJM chain, rate-base path, AEP/Virginia worksheets.
Module C. Local Public Ledger	County, water, roads, schools, air, noise, land use, and local public finance.	Tax/water/roads/air templates, local NPV worksheet, later-user credit rule.
Module D. Docket Package	Turning the method into filings, questions, orders, and reviewable conditions.	Data requests, protective-order rules, Ledger Adequacy Certificate, sample order language, full hypothetical worksheet.

Front-Door Packet

Use these five items before reading the longer theory and case discussion.

Item	What to fill or demand first
Minimum viable review	MW, service date, ramp schedule, legal payer, parent support, direct cost, shared cost, minimum bill, exit fee, RTO exposure, local exposure, residual claimant.
Ledger Adequacy Certificate	Asset, cost item, amount, unit, payer, instrument, collateral, evidence status, jurisdiction, residual claimant, and gap under stress.
Sample order conditions	Recommended public-cost-recovery defaults; no confidentiality without public representative review; no forecast inclusion without verifiable commitment; no flexibility credit without telemetry and penalties.
Hypothetical worksheet summary	Exposure, unit, instrument, credited value, gap, and residual claimant for direct facilities, shared network, underuse, exit, capacity/RBP, water, and local fiscal lines.
Evidence ladder	Strong evidence changes the ledger; proxy evidence points to likely risk; press releases show a story, not settlement.

How To Use This Toolkit

This is the technical companion to the shorter public brief, *Clearing the Ledger: Who Pays for AI Data Center Infrastructure?* The public brief carries the argument. This toolkit carries the working machinery: valuation haircuts, ledger scorecards, stress scenarios, legal hooks, PJM settlement-chain questions, technical-performance data requests, local fiscal templates, evidence ladders, and case worksheets.

For Liquidity Happy Hours, this toolkit should not be the first public doorway. It is the research engine behind a layered civic product: a hearing card for residents, a reporter kit for evidence and misuse warnings, this toolkit for advocates and regulators, and jurisdiction-specific legal appendices for filings.

Public launch MVP link

The RC3 public package is organized as a product ladder: Pocket Card for residents, Lite Ledger Review for action-oriented screening, Claim-to-Ledger Analyzer for turning public claims into evidence requests, County X for a full walkthrough, Reporter Kit for source discipline, and this toolkit for the research desk. The toolkit should answer expert questions after the public tools have opened the door.

Public product ladder

- **Civic handout:** five questions, pass card, fail card, document request, and 90-second hearing script.
- **Reporter kit:** evidence ladder, source-pinning register, PJM do-not-pro-rate warning, and three common misreports.
- **Advocate / PUC toolkit:** Ledger Adequacy Test, stress cases, certificate, jurisdiction map, and sample order conditions.
- **Legal appendix:** state-specific statutes, tariff sheets, docket numbers, order paragraphs, protective-order rules, and counsel review.

It is designed for public utility commission staff, public counsel, consumer advocates, local officials, energy reporters, community organizations, and researchers who need to turn a public promise into a docket-ready question. A reader does not need to use every table. The sequence is:

1. Start with the one-page ledger test: identify the asset, the bill, and the residual claimant.
2. Use the scorecards only for projects that cross the materiality triggers: large new load, public rate recovery, tax incentives, utility-owned facilities, RTO forecast inclusion, or thin project-company credit.
3. Keep ledgers separate unless an enforceable transfer mechanism links them. A county tax benefit does not erase a residential electric-rate exposure. A water clawback does not cover a PJM capacity charge.
4. Treat haircuts as an evidentiary discipline, not as an actuarial pricing model. They prevent weak promises from being counted at face value; they do not by themselves estimate default probability, bankruptcy recovery, or capacity-market causation.

Table 1: User pathway: which tools to use first

Reader	Start here	Then use
Resident or reporter	Minimum viable ledger review and evidence ladder	Payment-chain diagram, outer public ledger, hidden-ledger rules, and case snapshots.
PUC staff	Ledger Adequacy Test and valuation haircuts	Stress scenarios, ratepayer-protection matrix, sample order conditions, and docket scorecard.
County or city official	Local fiscal/environmental ledger	Tax, water, roads, school, land-use, and clawback NPV templates.
Public counsel or consumer advocate	Legal hooks and protective-order rules	Data requests, redacted-contract review, worked example, and Ledger Adequacy Certificate.

Minimum viable ledger review: ten questions

A reader with only one hour should answer: project MW; service date and ramp schedule; legal payer; parent support; direct interconnection cost and payer; shared network costs and assignment; minimum bill and billing determinant; exit fee versus remaining net book value; RTO or capacity exposure and pass-through; local fiscal, water, road, air, and residual claimant exposure.

Three-tier burden rule

1. **Ordinary commercial service.** If a project does not seek public subsidy, public infrastructure, rate-base recovery, RTO forecast treatment, public debt, or expedited approval that shifts risk, ordinary tariff and interconnection review may be enough.
2. **Public-risk trigger.** Tax incentives, utility rate recovery, RTO or IRP forecast inclusion, public-bond-backed infrastructure, water-authority debt, or expedited approval with risk shifting should trigger a Ledger Disclosure Duty.
3. **Public-cost transfer.** If reviewable evidence is missing, the recommended consequence is not a finding that the project is unlawful or should not be built. It is that unpriced risk should not be shifted to non-consenting public residual claimants.

Table 2: Default materiality trigger menu

Trigger	Default screening threshold to adapt by jurisdiction
MW threshold	25, 50, or 100 MW, or the relevant state large-load tariff threshold.
System peak share	Load large enough to affect utility, LSE, zone, or resource-plan peak assumptions.
Capex threshold	Customer-specific or shared grid, water, road, or public-safety capex above the local filing or board-approval threshold.
Forecast trigger	Entry into an IRP, RTO forecast, utility load forecast, capacity auction, or water-authority demand forecast.
Public support trigger	Tax abatement, PILOT, public bond, public infrastructure, expedited approval, or rate/rider recovery.
Credit trigger	Thin SPV, affiliate split, colocation lease, unidentified legal payer, or non-parent project company.
Local trigger	Water capacity reservation, backup-generation permit, major road/fire/school/public-safety cost, or concentrated land-use burden.

Module map

Use this toolkit as four drawers, not as a linear essay. Module A is screening: materiality triggers, taxonomy, proof levels, and evidence ladder. Module B is the electric and RTO ledger: adequacy test, valuation haircuts, stress scenarios, PJM chain, and rate-base path. Module C is the local ledger: tax, water, roads, air, schools, land, and NPV templates. Module D is the docket package: data requests, protective-order rules, Ledger Adequacy Certificate, worked example, and sample order language.

Task-first directory

Module A answers: should this project receive full review? Module B answers: who pays electric, transmission, capacity, and rate-base costs? Module C answers: who pays water, roads, schools, air, noise, and local debt? Module D answers: what should staff, public counsel, or a commission order require before public-cost recovery is allowed?

Table 3: Which pathway matters where?

Region or model	Main risk path	Use which module
PJM or capacity-market regions	RPM, Reliability Backstop Procurement, capacity pass-through, LSE allocation.	Module B: market-pooling ledger and PJM settlement chain.
Vertically integrated Southeast and similar states	IRP forecast, generation buildout, transmission expansion, rate base, riders.	Module B: rate-base expansion ledger and stress scenarios.
Public power or cooperative territory	Local board approval, wholesale contract, debt-backed facilities, member rates.	Modules A and C: screening plus local public ledger.
Water-stressed West	Cooling water, land, transmission siting, drought-year constraints.	Module C: local fiscal and environmental ledger.
Behind-the-meter or co-located sites	Standby service, deliverability, emissions, emergency curtailment, backup generation.	Modules B and C: technical performance plus local ledger.

What this toolkit is not

This toolkit does not decide whether an AI data center should be built. It asks whether public balance sheets may be used before the payment chain is clear. Its decision rules are proposed rules for public-cost recovery and public support, not a claim that every institution already has full jurisdiction over every ledger.

Where the framework can overstate risk

The ledger can overstate cost-shift risk if it misassigns shared infrastructure, ignores high-load-factor system benefits, treats ordinary system needs as project-caused, or discounts strong legal instruments too heavily. The remedy is not to abandon the test; it is to require costs and benefits to pass the same causation, evidence, and stress standards.

How this toolkit can be abused

Opponents can abuse the toolkit by calling every unknown line a subsidy. Supporters can abuse it by calling one clean ledger a clean project. Media users can abuse it by turning PJM market-level counterfactuals into project bills. Local governments can abuse it by treating tax revenue as electric-rate protection. Companies can abuse it by treating press releases, pledges, or voluntary flexibility claims as settlement resources.

Executive Summary

A county welcomes a billion-dollar AI data center. Officials promise jobs and tax revenue. The company promises innovation. The utility says the grid may need new equipment. Then residents ask the question that cuts through the press releases: will this show up on my electric bill?

That is the right question. But it is only the beginning. The deeper public question is:

The public question

Who receives the benefits, who signs the contracts, who finances the grid upgrades, and who pays if the project does not perform as promised?

This brief uses a Money View approach. In plain English, that means asking who owns the asset, who owes the money, and who must produce cash when the promise comes due. A data center project creates assets for some parties and payment obligations for others. Before the debate turns into slogans, the public deserves to see the contracts.

The core finding is simple: the AI data center debate is not just about how much electricity data centers use. It is about who pays for power capacity, transmission upgrades, backup resources, tax breaks, water infrastructure, and stranded assets if demand forecasts turn out to be wrong.

This brief focuses on the United States and uses real regulatory examples from Ohio, Virginia, and PJM. It also distinguishes between market-pooling regions and traditionally regulated, vertically integrated utilities. Those cases show that the question is not theoretical. Regulators are already writing new rules to decide whether large data centers, utilities, regional markets, or ordinary customers carry the risk.

This version keeps the topic narrow. The brief is about one public problem: how to read the American data-center buildout through contracts, tariffs, capacity markets, and household bills.

It also keeps the originality claim modest. This brief does not discover the ratepayer cost-shifting problem, invent large-load tariff tools, or offer a complete economic model. It offers an audit protocol for converting public promises into enforceable settlement resources, then testing whether those resources cover project-caused downside exposure across electric, regional-market, fiscal, water, technical-performance, and credit ledgers.

Burden of proof

When a project asks for public infrastructure, rate recovery, tax incentives, or expedited permitting, the burden should not fall on residents to prove a hidden subsidy. The applicant, utility, and public agency should show the tariff, contract, budget, or regulatory order that prevents cost spillover.

The supporter countercase

A data center is not automatically a public subsidy. The framework is not a presumption of guilt; it is a burden-of-proof rule. Supporters should win when their benefits settle. If the tariff, cost-to-serve study, load profile, credit support, regional allocation, and local fiscal ledger show that the project covers incremental costs and contributes surplus margin, the project should pass the ledger test.

Anti-misuse rule

Do not claim subsidy merely because a ledger line is unknown. Do not claim that a project is clean merely because one ledger passes. The correct conclusion is ledger-specific: passed, conditional, unresolved, failed under stress, not applicable, not triggered, or public-welfare override.

How to read this technical companion

The companion public brief carries the narrative argument: asset, bill, backstop, and payment chain. This document is the toolkit: valuation haircuts, stress scenarios, certificates, docket questions, and local NPV templates. It is intentionally denser than the public brief so the method can be tested in regulatory and public-policy settings.

1 What This Brief Adds, and What It Does Not

Several important reports have already mapped the data-center ratepayer problem. Harvard's Electricity Law Initiative explains how special contracts, confidential treatment, transmission cost allocation, and cost-causation disputes can shift Big Tech's power costs to the public. DOE and Berkeley Lab identify large-load rate-design tools such as minimum charges, study fees, collateral, credit requirements, and contract-size rules. RAP emphasizes transparency, planning, scenarios, sensitivities, and load flexibility. CoBank tracks the emerging tariff archetype across utilities and states. E3's Amazon-commissioned ratepayer study presents the strongest countercase: under some existing tariffs and facility-level assumptions, data centers can generate enough utility revenue to cover cost to serve and may even create surplus revenue for other customers.

This brief builds on that work. It does not claim that minimum bills, exit fees, collateral, separate rate classes, or flexible-load requirements are new. The claim is narrower: Money View gives public readers a disciplined way to connect those tools into one payment chain and then stress-test it.

Table 4: Where this brief fits in the existing debate

Existing work	What it already contributes	What this brief adds
Harvard Electricity Law Initiative	Shows how utility rate structures, special contracts, secrecy, and cost-causation disputes can transfer data-center costs to ratepayers.	Reframes those transfers as balance-sheet links and asks where the payment chain fails under stress.
DOE / Berkeley Lab large-load rate design	Catalogs tariff elements such as study costs, minimum demand charges, credit tests, collateral, contract size, and term length.	Treats each tariff element as a settlement device: what promise does it secure, and how much loss does it actually cover?
RAP large-load planning work	Emphasizes transparency, interagency coordination, planning, scenarios, sensitivities, and flexible load.	Turns planning uncertainty into a ledger question: who pays if the forecast used in planning is wrong?
PJM and Monitoring Analytics	Provide market evidence that data-center load growth is affecting capacity-market conditions.	Separates market-pooling spillovers from vertically integrated rate-base expansion.
AEP Ohio and Virginia SCC	Provide real tariff examples with minimum bills, collateral, separate rate classes, and long service terms.	Stress-tests whether those provisions are sufficient under cancellation, underuse, credit deterioration, and regional market spillovers.
White House Ratepayer Protection Pledge	Provides a national political commitment that hyperscalers and AI companies will build, bring, or buy energy and pay for related infrastructure.	Asks whether the pledge becomes an enforceable tariff, service agreement, collateral package, or only a public promise.
E3 / Amazon ratepayer study	Argues that evaluated Amazon data centers produced sufficient or surplus utility revenue relative to cost to serve, creating possible downward rate pressure.	Treats that claim as a valid supporter countercase, but asks whether the same result survives local cost, regional capacity, forecast-error, credit, and fiscal stress tests.

This brief does not offer a new tariff device. It offers an audit protocol for converting public promises into enforceable settlement resources, then testing whether those resources cover project-caused downside exposure across electric, regional-market, fiscal, water, and credit ledgers.

2 Start With the Bill

Imagine a county where a large AI data center is proposed. Local officials say the project will bring investment, tax revenue, and a place in the next technology boom. The company says it will be a good neighbor. The electric utility says new substations or transmission upgrades may be needed. Residents ask whether their electric bills will rise.

At that point the debate usually splits into familiar camps. One side says, ”This is the future, and opposition is anti-technology.” The other says, ”This is a private project using public resources, and ordinary customers will pay.”

The Money View does not begin by choosing a side. It begins by asking three questions:

Three Money View questions

1. Who promises to pay?
2. Who receives the cash flow?
3. Whose balance sheet records the new asset, and whose balance sheet records the new liability?

These questions bring the debate back to the ledger. A data center is not just a building full of servers. It is a chain of future payment promises: for land, power, cooling, transmission, backup capacity, taxes, water, roads, and public services. Each promise must eventually be settled.

3 A Plain-English Toolkit

This brief uses four core terms. Everything else is explained as it appears.

Table 5: Key terms in plain English

Term	Plain-English meaning
Balance sheet	A record of what an entity owns, what it owes, and how future cash flows enter or leave that entity.
Balance sheet graph	A map showing how several balance sheets are connected through contracts, bills, subsidies, loans, and public obligations.
Sources-and-uses account	A table showing where project money comes from and what it is used for. In Money View notes this is sometimes called a use-source account.
Rate base	The pool of utility assets that regulators allow a utility to recover through customer bills, usually with an allowed return.

Two other ideas will appear in ordinary language. A *stranded asset* is a grid or power investment built for demand that never arrives. A *tax abatement* is a local tax break offered to attract investment.

4 Data Center Risk Taxonomy

Not every data center creates the same public risk. A hyperscaler with an investment-grade parent guarantee is different from a speculative developer using a thin project company. A flexible load that can curtail during peak stress is different from a rigid 24/7 firm load. A project that pays for its own substation can still create regional capacity costs, while a project with strong electric protections can still receive local tax or water subsidies.

Table 6: Project type changes payment-chain risk

Project type	Main payment-chain risk	What the public should ask
Hyperscaler owned and operated	The company may have deep credit, but the project can still create shared grid, capacity, water, and fiscal costs.	Is the parent guarantee unconditional, and does it cover shared costs, not only direct interconnection?
Hyperscaler leasing from a colocation provider	The brand name may be strong while the legal payer is a landlord, affiliate, or project company.	Who signs the service agreement, and who pays if the tenant leaves?
Speculative developer or queue holder	The project may reserve grid planning capacity before a committed end user exists.	Is there a study fee, deposit, milestone schedule, and cancellation reimbursement?
Behind-the-meter or co-located load	The project may claim to avoid grid costs while still relying on standby service, transmission, or capacity markets.	What standby, backup, deliverability, and reliability obligations remain on the grid?
Firm 24/7 load	The grid must reserve capacity at the hardest hours, even if annual energy is matched with clean-energy purchases.	What minimum bill, capacity charge, or resource contribution pays for that reservation?
Flexible or interruptible load	The project may reduce system stress if curtailment is real, measurable, and enforceable.	What performance penalty applies if the load fails to curtail during grid-stress hours?
Thin SPV or affiliate structure	The asset may be valuable while the legal payer has limited assets at default.	Is collateral senior, liquid, and callable before bankruptcy or restructuring traps recovery?

This taxonomy keeps the argument from becoming a trial of ”data centers” as one uniform category. A project can pass the ledger test if its particular risk structure is matched with enforceable payment obligations.

4.1 Why AI data centers are not just another large load

Much of this toolkit also applies to semiconductor fabs, hydrogen electrolyzers, EV factories, LNG facilities, crypto mining, and industrial parks. AI data centers deserve special treatment because their risks combine in a distinctive way:

- **Scale and speed:** load requests can grow faster than ordinary IRP and transmission-planning cycles.
- **Geographic clustering:** campuses can concentrate around specific substations, zones, water systems, and land-use corridors.
- **Entity complexity:** the public brand may be a hyperscaler while the legal payer is a colocation landlord, affiliate, SPV, or phased campus entity.
- **Demand uncertainty:** training, inference, and cloud demand can change with AI business models and hardware cycles.
- **Opaque flexibility:** outside parties often cannot tell whether workloads are actually curtailable without telemetry, scheduler evidence, and contract rights.
- **Water-power tradeoff:** cooling choices can shift stress between electricity peaks and water systems.
- **Polarized public narrative:** the same project may be framed as national competitiveness, local jobs, environmental threat, or corporate extraction.

Table 7: Technical-economic distinctions that change the ledger

Distinction	Why it changes cost causation	Ledger implication
Training cluster vs. inference or data-serving load	Training may be more shiftable if workloads can pause; inference may need low-latency, always-on service.	Do not value curtailment until the workload type and service-level commitments are known.
GPU-intensive AI campus vs. ordinary cloud expansion	A concentrated GPU campus can drive sharper ramp, cooling, and local capacity needs.	Require project-specific ramp schedule, power-quality standards, and cooling assumptions.
Firm load vs. curtailable load vs. emergency-only curtailment	Emergency-only curtailment may help reliability but may not reduce ordinary capacity obligations.	Apply a haircut to flexibility value unless curtailment is dispatchable before scarcity costs are incurred.
Annual renewable matching vs. hourly deliverable clean capacity	Annual matching can clean the energy story while leaving peak-hour reliability unresolved.	Count clean-energy contracts only to the extent they are deliverable at constrained hours.
Behind-the-meter gas, nuclear, or storage vs. grid-dependent backup	Onsite resources can reduce grid exposure or create standby, fuel, emissions, and failure risks.	Require standby tariffs, deliverability tests, emissions permits, and failure penalties.
Single campus vs. phased affiliate buildout	Phasing can hide total MW below thresholds or move risk across affiliates.	Aggregate affiliated phases and require milestone deposits tied to the full campus plan.
Load factor certainty: 95, 80, or 60 percent	Lower realized load can leave fixed infrastructure costs unrecovered.	Stress-test minimum bills and exit fees against low-load and delayed-ramp cases.

Table 8: From project type to default regulatory response

Project type	Default regulatory response
Hyperscaler owned and operated	Require an unconditional parent guarantee, coverage for shared grid costs, and disclosure of local fiscal terms.
Hyperscaler leasing from a colocation provider	Require a tenant-departure clause, landlord credit support, assignment restrictions, and notice if the named tenant changes.
Speculative developer or queue holder	Require milestone deposits, queue-withdrawal penalties, study-cost reimbursement, and cancellation payment for committed buildout costs.
Behind-the-meter or co-located load	Require a standby tariff, deliverability test, emissions permits, failure penalties, and clear rules for when the project still relies on the grid.
Flexible or interruptible load	Require verified curtailment, telemetry, audit rights, performance penalties, and a haircut before any capacity credit is counted.
Thin SPV or affiliate structure	Require cash collateral, a letter of credit, escrowed funds, a larger exit fee, or a parent guarantee that reaches a deep-credit entity.

5 Why This Is a Money View, Not Just a Spreadsheet

It is fair to ask whether this is just corporate finance with a new label. A simple accounting report would stop at sources, uses, assets, and liabilities. Those tools are useful, but they are only the doorway.

The Money View adds a harder question: what happens at settlement? When the project has to take power, pay bills, finance upgrades, and survive a forecast error, who has the liquidity to keep the chain from breaking?

Perry Mehrling's Money View emphasizes survival constraints, the hierarchy of money, and the institutions that make markets by standing between buyers and sellers. In finance, a dealer keeps markets going by absorbing inventory, funding positions, and finding the next buyer. The dealer's balance sheet becomes the bridge between today's trade and tomorrow's settlement.

In the data center setting, the key intermediaries are not Wall Street dealers. They are electric utilities, public utility commissions, regional grid operators, and capacity markets. They stand between the data center and the rest of the power system. They decide whether a large new load remains a private obligation or becomes part of a wider pool.

The analogy has limits. A franchised electric utility is not a free-trading Wall Street dealer. It is a regulated monopoly with a public-service duty, cost-of-service rules, commission oversight, and legally filed tariffs. That is why this brief uses a two-language bridge. Money View terms such as settlement pressure, survival constraint, and hierarchy of promises should be read alongside standard utility-regulation terms: cost causation, prudence review, used-and-useful assets, rate design, and stranded-cost protection.

Table 9: Money View terms translated into utility-regulation language

Money View idea	What it means for AI data centers	Utility-regulation counterpart
Settlement	A promise is real only when it can be paid on the date cash is due. A press release does not settle a power bill, a capacity charge, or a construction invoice.	Filed tariff, enforceable service agreement, commission order, and cost-recovery mechanism.
Survival constraint	If demand is delayed, costs rise, or the project is canceled, someone still has to survive the cash drain.	Stranded-asset risk, rate cross-subsidization risk, prudence review, and ring-fencing of large-load costs.
Hierarchy of money	Some promises are stronger than others. Cash, collateral, minimum bills, and signed tariffs sit higher in the hierarchy than informal assurances.	Tariff rate design and enforceable collateral architecture, including letters of credit, parent guarantees, deposits, and minimum bills.
Dealer or market-maker role	The utility and regional grid operator transform one customer's load request into grid investment, market forecasts, and system-wide charges.	Regulated load-serving intermediary under cost-causation rules, RTO market rules, and public utility commission oversight.

That is why the balance sheet graph matters. It is not just a static accounting diagram. It shows where settlement pressure may move if the data center uses less power than promised, if grid upgrades cost more than expected, or if capacity prices rise for everyone.

The extra analytical value, if there is one, is not that Money View replaces utility regulation. It is that it forces four dockets that are often separated–utility rate cases, RTO markets, local fiscal incentives, and corporate finance–onto the same page.

Table 10: Where the Money View should add discipline

Ordinary regulatory question	Money View follow-up	Why the follow-up matters
Is the tariff just and reasonable?	If the customer underuses or exits, which balance sheet absorbs the cash shortfall?	A tariff can look fair at signing but fail under nonperformance.
Does the asset pass prudence review?	Who funded construction before recovery, and who supplies liquidity while recovery is delayed?	Timing matters; a solvent project can still create cash-flow stress elsewhere.
Does the contract include collateral?	Is the collateral senior, liquid, and available before bankruptcy or affiliate restructuring blocks recovery?	A weak guarantee can sit below other claims in the hierarchy of promises.
Does the project bring tax revenue?	Which public entity gave up revenue, built infrastructure, or accepted environmental risk?	The electric ledger and local fiscal ledger can point in opposite directions.

5.1 Where Money View changes the answer

The test for this framework is whether it changes a regulatory judgment, not merely the vocabulary. The following are the cases where it should bite.

Table 11: Where the Money View changes the answer

Scenario	Ordinary answer	Money View answer
State tariff looks protective, but PJM capacity costs remain pooled	The PUC tariff passes because direct interconnection costs and minimum bills are assigned to the large-load customer.	The state ledger passes, but the regional capacity ledger may fail. The question moves to PJM, FERC, the LSE, and the retail pass-through mechanism.
County fiscal note is positive, but utility rate base contains unring-fenced assets	The economic-development deal looks beneficial because local tax revenue after abatements is positive.	The local fiscal ledger may pass while the electric ledger fails. A county win can coexist with household electric-bill exposure.
Parent guarantee exists, but only covers direct project bills	The company looks creditworthy because a deep parent stands behind the service agreement.	The credit ledger passes only for the covered obligation. Shared network costs, capacity exposure, water debt, or tax concessions can still remain outside the guarantee.

Table 12: Where the ledger changes the legal or regulatory outcome

If the review stops here	Ordinary outcome	Ledger outcome
Retail tariff only	Approve because direct facilities, minimum bills, and collateral exist.	Conditional approval: regional capacity and transmission pass-through exposure remain unproved.
County fiscal note only	Support because tax revenue after abatement is positive.	Require local statement plus electric/RTO statement; fiscal surplus does not settle household bill exposure.
Parent guarantee only	Treat credit risk as solved because the corporate parent is strong.	Count guarantee only for obligations inside its legal scope; shared network, water, and capacity exposure may still need instruments.
Opposition claim only	Treat the project as a subsidy because it is large and politically unpopular.	Project can pass if cost-to-serve surplus, later-user credits, accredited flexibility, and local NPV survive stress.
Confidential contract only	Accept sealed terms as sufficient because utility and customer negotiated them.	Allow confidentiality only with public representative review and a public non-sensitive summary of payer, term, collateral, cost categories, and residual claimant.

That is the conceptual bite. Money View does not say cost causation is wrong. It says cost causation has to settle across every ledger where the project creates claims. A project can pass one docket and still fail the payment chain.

The Money View test

A claim is not settled when a company says ”we will pay.” It is settled when contracts, tariffs, collateral, minimum bills, and exit fees show who must provide cash if the forecast fails.

Three Money View propositions

1. **Utility/RTO as settlement intermediary.** Large-load requests become public risk when utilities or RTOs transform private load promises into system planning, infrastructure finance, or pooled market obligations.
2. **Public-risk shifting as unmatched claim creation.** Cost shifting occurs when a public ledger receives an obligation without a same-ledger enforceable settlement resource.
3. **Adequacy depends on hierarchy, timing, jurisdiction, and stress survival.** A promise protects the public only if it is senior enough, liquid enough, timely enough, jurisdiction-matched,

and durable under delay, underuse, exit, default, curtailment failure, or forecast error.

6 Causation Gate: Before Running the Ledger

The Ledger Adequacy Test should not label every system investment as data-center-caused. **Project-caused exposure** means incremental cost, acceleration, risk, or pooled-market exposure that the record ties to the project. It does not mean every system need near the project. Before calculating downside exposure, classify each cost by causation type.

Table 13: Causation gate for project-caused exposure

Causation category	Test	Default ledger treatment
But-for direct cost	The facility, study, interconnection, or service would not be built but for this project.	Direct assignment to the project through upfront contribution, tariff charge, reimbursement, or service agreement.
Triggered shared cost	The project triggers construction, but later users or the broader system may benefit.	Initial assignment to triggering project, with later-user credit, refund, reassignment, or beneficiary sharing.
Forecast-reliance cost	The project enters utility, IRP, RTO, water, or local fiscal forecasts and causes earlier procurement, construction, or debt issuance.	Require milestone schedule, deposit, cancellation payment, exit protection, and forecast-error sharing.
Pooled market exposure	The project contributes to a regional market or pooled cost, but cannot be mechanically assigned dollar-for-dollar.	Make a spillover-risk finding; do not impose project-level dollar liability without a legal assignment mechanism.
General system need	Retirements, electrification, ordinary growth, reliability standards, or broad public policy would require the investment even without the project.	Do not assign to the project unless the record shows incremental cost, acceleration, or customer-specific exposure.

Causation before arithmetic

First classify the cost, then run the ledger. Otherwise the framework can over-assign ordinary system needs to AI data centers, or under-assign forecast-reliance and shared costs that the project actually triggered.

Table 14: Four-stage ledger line test

Stage	Question	Required output
Engineering or planning causation	Did the project trigger, accelerate, enlarge, or make more uncertain the cost, procurement, debt, or forecast?	But-for direct, triggered shared, forecast-reliance, pooled market, or general system need.
Regulatory assignment	Under current tariff, RTO rule, PUC order, local agreement, bond covenant, or permit, who is assigned the cost today?	Current payer, customer class, LSE/EDC, utility shareholder, taxpayer, water user, or unresolved pool.
Settlement instrument	Which tariff, contract, collateral, minimum bill, exit fee, guarantee, PILOT, water rate, or rider produces cash or performance?	Credited settlement resource, unit, timing, scope, evidence status, and residual claimant.
Remedy authority	Which institution can order, condition, disclose, review under protective order, or refer the unresolved line elsewhere?	PUC, county, water authority, RTO/FERC, environmental agency, public counsel, or court.

Table 15: Causation Evidence Matrix: strong and medium evidence

Cost item	Strong evidence	Medium evidence
Direct interconnection	Executed interconnection agreement, filed tariff charge, or service agreement naming facilities and payer.	Utility testimony, signed study result, or commission staff report assigning customer-specific facilities.
Shared network upgrade	Facility study with beneficiary allocation, later-user refund rule, or commission order assigning cost shares.	Utility testimony identifying project-triggered upgrade and future users.
Forecast-reliance cost	IRP/RTO/utility scenario with and without the project, executed customer commitment, milestone, and forecast entry.	Single load-growth assumption tied to named project or signed queue milestone.
Water infrastructure	Bond disclosure, water-service agreement, or authority order naming project demand and debt-service payer.	Water authority minutes or staff memo tying capacity expansion to project load.
Road, fire, or public-safety cost	Development agreement, capital plan, or budget line assigning project-specific cost.	Agency memo estimating staffing, road, or emergency-service needs.
School or local fiscal effect	Signed abatement, PILOT, fiscal note, school allocation, and clawback terms.	Economic-development memo with tax assumptions and school split.
Permit-governed local burden	Air, water, land-use, or backup-generation permit with operating limits, monitoring, and enforcement.	Draft permit, staff report, or mitigation plan.

Table 16: Causation Evidence Matrix: proxy and unknown evidence

Cost item	Proxy evidence	Unknown or weak evidence
Direct interconnection	Public announcement naming substation or feeder need.	Press release or vague grid-upgrade claim.
Shared network upgrade	General capex plan with location and timing consistent with project.	Broad system upgrade with no project link.
Forecast-reliance cost	Queue position, investor call, or developer presentation.	Rumor, duplicate queue request, or speculative headline MW.
Water infrastructure	Developer presentation or local news about water need.	General water-system plan with no project-specific demand.
Road, fire, or public-safety cost	Hearing testimony or planning-board discussion.	Headline investment or jobs number only.
School or local fiscal effect	Public official statement about expected tax benefit.	Gross investment number with no abatement or school analysis.
Permit-governed local burden	Site plan or public presentation.	General assurance that impacts are minimal.

7 Cluster and Queue Ledger: When the Project Is Not the Unit

AI data-center risk often arrives as a cluster rather than as one clean project. A single campus may look manageable while an affiliated buildout, colocation queue, or zone-level forecast changes the marginal cost of capacity, transmission, water, or local infrastructure. The ledger should therefore test both the individual project and the relevant cluster.

Table 17: Cluster and queue ledger

Cluster question	Evidence to request	Ledger treatment
Territory aggregation	All pending and executed large-load requests in the same utility territory, zone, water system, or substation pocket.	Run materiality triggers at the cluster level, not only project by project.
Phased affiliates	Common ownership, affiliate structure, adjacent parcels, shared substation, shared water agreement, or common parent support.	Aggregate phased affiliates when they would evade MW, credit, or minimum-bill thresholds if filed separately.
Speculative queue de-duplication	End-user identity, tenant commitment, deposits, milestone schedule, and duplicate applications across utilities or zones.	Discount or exclude speculative requests from forecast reliance unless backed by verifiable commitment.
Acceleration of shared upgrades	Planning record showing whether the project or cluster moves a transmission, generation, substation, road, or water project forward by 3–5 years.	Treat acceleration as forecast-reliance or triggered shared cost, with later-user credit if future beneficiaries materialize.
Trigger point	Facility study, IRP scenario, RTO forecast, water-authority plan, or local capital plan showing whether the trigger is a single project, a project portfolio, or a forecast class.	Assign direct costs to direct triggers; use beneficiary allocation, forecast-error sharing, or public-welfare finding for cluster-triggered costs.

Do not let the project boundary hide the queue

A project-level ledger is necessary but not sufficient where multiple affiliated or queued loads create the marginal need. If the risk is caused by a cluster, the residual claimant analysis should name the cluster, queue, zone, or forecast class that created it.

Table 18: Cluster cost allocation rule

Allocation case	When to use it	Default treatment
Direct trigger	One project alone triggers customer-specific facilities or a clearly separable upgrade.	Direct assignment to that project through contribution, tariff, reimbursement, or service agreement.
Proportional MW or peak contribution	Multiple committed projects jointly drive a shared upgrade and coincident peak contribution or contracted MW is measurable.	Allocate by coincident peak, contracted MW, ramp schedule, or another filed allocator; disclose allocator and sensitivity.
Acceleration share	The project or cluster pulls forward an otherwise-needed public investment by a defined number of years.	Assign only the acceleration value, carrying cost, or early-procurement premium unless the record shows exclusive benefit.
Portfolio trigger	A class of large-load projects, affiliated phases, or queue portfolio triggers the cost.	Use class-level rider, cluster deposit pool, beneficiary allocation, later-user refund, or portfolio true-up.
Unassignable pooled exposure	The record shows regional spillover risk but not a legally defensible project-level allocator.	Make a spillover-risk finding; do not convert it into a project bill without RTO, state, contract, or welfare-finding authority.
General system need	Ordinary load growth, retirements, electrification, and reliability standards would require the cost without the cluster.	Do not assign to data centers except for incremental cost, acceleration, or identifiable forecast-risk premium.

Table 19: Worked allocation example for a shared cluster upgrade

Record finding	Example fact pattern	Allocation method	Ledger result
Direct trigger	One 300 MW campus requires a dedicated substation with no later users.	Direct assignment through contribution, tariff, or service agreement.	Project-level electric ledger line.
Portfolio trigger	Five committed projects reserve 1 GW in the same substation pocket and jointly trigger a transformer bank.	Allocate by coincident peak, reserved capacity, or filed class allocator; true up when phases energize.	Cluster deposit pool or class rider with later-user refund.
Acceleration value	Transmission upgrade was planned for 2032, but the cluster moves need to 2028.	Assign four years of acceleration carrying cost or early-procurement premium, not the full asset, unless exclusive benefit is shown.	Forecast-reliance ledger line, with future-beneficiary credit.
Ordinary system need	Retirements and general load growth would require the upgrade by the same date without the data-center queue.	Do not assign the full cost to data centers; disclose any incremental scope or risk premium.	General system ledger, not project-caused exposure.
Unassignable market effect	Data-center forecast contributes to regional capacity price pressure, but no defensible customer allocator exists.	Make a spillover-risk finding; require RTO/state remedy before project-level charge.	Regional ledger unresolved, not a project bill.

Acceleration-value formula

When a data-center cluster accelerates an otherwise-needed asset, do not automatically allocate the full asset cost to the last project in the queue. A screening formula is:

$$\text{Acceleration exposure} = \left(\frac{C}{(1+r)^{t_{\text{new}}}} - \frac{C}{(1+r)^{t_{\text{old}}}} \right) \times s.$$

Here, s is the project's reserved-MW, coincident-peak, UCAP, or other filed allocator share of the committed cluster. Example: a 500 MW committed cluster includes a 200 MW project. The cluster moves a \$300 million transmission upgrade from year 7 to year 2. At a 7 percent discount rate, the acceleration value is roughly \$75 million; the 200 MW project's proportional screening share is about \$30 million before later-user credits, true-ups, or welfare findings. The project-specific exposure is not the full \$300 million unless the record shows exclusive benefit or direct causation.

Screening allocation is not adjudicative allocation

Cluster formulas are screening tools. They identify the residual claimant, the rough scale of exposure, and the evidence that should be requested. They should not be treated as final bills unless replaced or confirmed by a filed allocator, beneficiary analysis, facility study, IRP or RTO scenario, settlement agreement, or commission order. A screening share asks what needs review; an adjudicative share decides who pays.

8 The Ledger Adequacy Test

To make the payment-chain idea usable, the brief uses a simple test. A project has adequate cost protection only if enforceable settlement resources are at least as large as the project-caused costs and downside exposures they are supposed to cover.

Ledger Adequacy Test

$$\begin{aligned}
\text{Settlement resources} = {} & \text{Upfront contributions} \\
& + \text{NPV of enforceable fixed payments} \\
& + \text{callable collateral and parent guarantees} \\
& + \text{exit fees and cancellation payments} \\
& + \text{enforceable capacity or resource contributions} \\
& + \text{discounted verified flexibility value} \\
& + \text{local fiscal clawbacks}
\end{aligned}$$

$$\begin{aligned}
\text{Downside exposure} = {} & \text{customer-specific grid costs} \\
& + \text{assigned shared network costs} \\
& + \text{forecast-error and stranded-asset risk} \\
& + \text{allocated regional capacity impact} \\
& + \text{local public-infrastructure and fiscal costs}
\end{aligned}$$

$$\text{Ledger gap} = \text{Settlement resources} - \text{Downside exposure}$$

If the ledger gap is positive under realistic downside scenarios, the project may be paying its own way. If the gap is negative, the missing dollars must land somewhere: the utility, its shareholders, other ratepayers, taxpayers, creditors, or future customers.

This is not a full financial model. It is a public test. It tells readers what numbers must be produced before a slogan such as ”Big Tech will pay” or ”households will be stuck with the bill” becomes a defensible claim.

A ledger is adequate only when the credited value of enforceable settlement resources covers the stressed value of project-caused public exposure in the same jurisdictional ledger. A county clawback does not cover PJM capacity exposure. A parent guarantee for direct electric bills does not cover water debt. A minimum bill does not cover remaining net book value unless its NPV is large enough under the exit year being tested.

Run the test ledger by ledger

The formula must be run ledger by ledger. A single project may have a positive electric ledger and a negative water ledger, or a positive county fiscal ledger and an unresolved PJM ledger. Aggregate only where an enforceable transfer mechanism links the ledgers.

8.1 Unit and NPV protocol

The formula is useful only if every ledger line states its unit, time horizon, and conversion rule. Do not add annual revenue shortfalls, one-time capital costs, MW obligations, water bonds, and non-monetized environmental burdens as if they were the same object.

Table 20: Unit normalization protocol

Field	Required entry	Why it matters
Unit	One-time dollars, dollars per year, NPV, MW, UCAP MW, MW-day, MWh, delivery-year obligation, bond-term exposure, or non-monetized exposure.	Prevents apples, substations, capacity obligations, and water debt from being collapsed into one number.
Time horizon	Construction period, service term, 5 years, 10 years, 20 years, asset life, bond term, delivery year, or abatement term.	A 36-month exit fee and a 40-year asset life are not comparable without time normalization.
Discount rate	Utility carrying cost, social discount rate, customer credit-adjusted rate, bond rate, or commission-approved rate.	NPV depends on whose cost of capital and risk is being measured.
Probability treatment	Base case, stress case, expected loss, or worst reasonable case.	A public screening score should not be confused with a formal expected-loss model.
Monetization status	Cash ledger, capacity ledger, environmental exposure ledger, or distributional/non-monetized finding.	Some harms should be disclosed and mitigated even if they are not reduced to dollars.
Evidence status	Public, confidential under protective order, proxy, missing, or contested.	Determines whether the line can support public-cost recovery.

Table 21: Decision rule for the Ledger Adequacy Test

Result	Recommended consequence
Positive under all required stress cases	Recommended finding for public-cost recovery: pass under the filed instruments, subject to ordinary review, reporting, and the institution's legal authority.
Positive under base case, negative under stress	Recommended conditional finding: require more collateral, higher minimum bills, longer take-or-pay terms, shareholder sharing, or a specific rider true-up before shifting risk to public balance sheets.
Negative under base case	Recommended default for public-cost recovery: no ordinary rate recovery, public subsidy, or public-risk shifting absent an adequate evidentiary showing, unless the decision-maker makes a supported finding that the cost is just, reasonable, and properly allocated.
Unknown because key documents are hidden	Recommended default for public-cost recovery: no ordinary rate recovery, public subsidy, or public-risk shifting absent an adequate evidentiary showing, unless staff, public counsel, or a consumer advocate can inspect the ledger under protective order.
Positive electric ledger, negative local ledger	Recommended: do not cite electric approval as proof that water, tax, road, school, public-safety, and environmental costs are settled.
Positive local ledger, negative regional-market ledger	Recommended: do not cite local approval as proof that PJM, RTO, or retail pass-through exposure has been settled.

Table 22: Layered adjudication rules: from audit to decision

Question	Standard	Recommended consequence
Is the project ordinary commercial service?	No public subsidy, no public-risk shifting, no special forecast reliance, and no unrecovered public infrastructure.	Apply ordinary tariff, interconnection, zoning, and environmental review.
Has a public-risk trigger occurred?	The project seeks rate recovery, public infrastructure, tax incentives, public debt, RTO/IRP forecast inclusion, or expedited approval that shifts risk.	Require ledger disclosure, public summary, and protective-order review for confidential terms.
Is the ledger line measurable?	Unit, time horizon, payer, instrument, jurisdiction, and residual claimant are identified.	Proceed to valuation; if not, classify as unknown and do not use it to justify public-cost recovery.
Is the settlement resource adequate under stress?	Credited value covers the same-ledger exposure under delay, underuse, exit, default, curtailment failure, or forecast error.	Pass, conditional pass, or require additional instrument before public-risk shifting.
Is a cross-ledger welfare tradeoff being claimed?	The decision-maker discloses winners, losers, distributional effects, compensation mechanism, and residual claimant.	Treat as an explicit public-welfare finding, not as ledger settlement.
Is aggregate settlement positive but burden concentrated?	Material costs fall on customers or communities that do not receive corresponding benefits, cannot exit, and were not represented in the negotiation.	Conditional pass with targeted mitigation, public-benefit agreement, hold-harmless payment, affordability fund, or explicit welfare finding.
Is procedural access inadequate?	Affected customers cannot inspect public summaries, public counsel lacks protective-order access, or notice/intervention barriers are material.	No final public-cost-recovery finding until public summary, representative review, or access remedy is provided where authorized.

Table 23: Ledger Certificate status taxonomy

Status	Meaning
Pass	Same-ledger settlement resources cover the tested exposure under required stress cases.
Conditional pass	The line can proceed only with additional collateral, true-up, mitigation, disclosure, or review.
Unresolved, evidence missing	Required evidence is absent from the public or protected record.
Unresolved, confidential but reviewable	Terms are sealed but available to staff, public counsel, or another representative under protective order.
Unresolved, contested	The parties dispute causation, amount, unit, assignment, or credited value.
Not applicable	The ledger line does not apply to this project or institution.
Not triggered	The line may matter later, but the triggering event has not occurred.
Public-welfare override	An authorized decision-maker accepts a cross-ledger tradeoff after naming winners, losers, compensation, distributional effects, and residual claimant.

Public-welfare override conditions

A public-welfare override is not a shortcut around settlement discipline. It should count only if: (1) an authorized public body makes the finding, not merely a staff memo, company press release, or economic-development announcement; (2) the record names winners, losers, compensation, distributional effects, and residual claimant; (3) the decision explains why same-ledger settlement is unavailable, unnecessary, or outweighed by a public purpose; and (4) the order includes a reopener, true-up, review date, or mitigation trigger.

Table 24: Two public-welfare override statuses

Override status	What it can do	What it cannot do
Policy-support override	Explain why an authorized public body supports the project despite an unresolved or cross-ledger burden.	It does not settle a payment obligation or prove that households, water users, schools, or retail classes are protected.
Payment-settlement override	Treat a burden as addressed because an appropriation, transfer, credit, rider, contract, mitigation fund, or other enforceable instrument moves value to the affected ledger.	It cannot rely only on jobs, innovation, AI leadership, or general economic-development narrative.

Technical flexibility belongs inside the test, not in a footnote. A data center that can curtail during grid-stress hours may reduce the capacity it causes the system to reserve. But that value should be discounted unless the curtailment is measurable, enforceable, and penalized if it fails.

Verified flexibility value

$$\begin{aligned}\text{Verified flexibility value} = {} & \text{expected avoided capacity or reliability cost} \\ & \times \text{performance probability} \\ & \times \text{enforceability factor} \\ & - \text{failure externality}\end{aligned}$$

If an interruptible commitment lacks telemetry, audit rights, event triggers, performance standards, and failure penalties, it should not be counted as a settlement resource.

8.2 Valuation rules: do not count weak promises at face value

The Ledger Adequacy Test can be manipulated if every promise is counted at face value. A parent guarantee, local clawback, or capacity contribution is valuable only to the extent that it produces usable cash or usable capacity in the ledger it is supposed to protect.

Credited settlement value

$$\begin{aligned}\text{Credited settlement value} = {} & \text{face value} \\ & \times \text{timing factor} \\ & \times \text{legal enforceability factor} \\ & \times \text{credit factor} \\ & \times \text{priority factor} \\ & \times \text{jurisdiction-match factor} \\ & \times \text{performance factor}\end{aligned}$$

These factors are not empirical estimates in this draft. They are default public-facing scores. In a docket, each should be replaced by instrument-specific evidence: governing law, call conditions, expiry, legal payer, credit rating, audited cash position, bankruptcy priority, and whether the instrument protects the correct ledger.

Table 25: Valuation rules for settlement resources

Discount	Question to ask	Why it matters
Timing discount	When does cash arrive: before construction, during operation, after default, or after litigation?	Cash before construction prevents risk; cash after bankruptcy may only reduce losses.
Legal discount	Is the promise a filed tariff, commission order, enforceable service agreement, private side letter, or press release?	A public statement should not be valued like a filed rate or secured contract.
Credit discount	Who pays: hyperscaler parent, affiliate, colocation landlord, thin SPV, or project tenant?	The payer's balance sheet determines whether the promise survives stress.
Priority discount	Is the claim senior, liquid, and callable before lenders, lessors, or secured creditors block recovery?	Face value can vanish if the utility stands behind stronger creditors.
Jurisdiction discount	Which ledger does the tool protect: electric rates, utility shareholders, local fiscal accounts, water debt, or regional capacity charges?	A county tax clawback may protect county revenue while leaving electric customers exposed.
Performance discount	For interruptibility or self-supply, is the resource deliverable at peak hours with measurement and penalties?	A capacity promise has little value if it fails when the grid needs it.

No double-counting rule

The same dollar cannot be counted twice. An upfront contribution cannot also be counted as minimum-bill recovery, exit-fee coverage, and local public benefit. Likewise, the same shared network upgrade should not be counted once as direct interconnection cost and again as regional capacity exposure unless the record shows two separate costs.

No cross-ledger offset without transfer

Settlement protection rule: a benefit in one ledger cannot erase a liability in another ledger unless an enforceable transfer mechanism moves cash, credit, capacity, or legal responsibility between them. County PILOT revenue does not cover PJM capacity charges; verified flexibility does not pay water debt; local jobs do not cover stranded rate-base assets.

Public welfare rule: a commission, county, or legislature may still make cross-ledger social-welfare tradeoffs. But it should do so explicitly by naming winners, losers, distributional effects, compensation mechanisms, and residual claimants. That is a public-welfare finding, not ledger settlement.

Utility protection is not always ratepayer protection

A settlement device counts as ratepayer protection only to the extent that it prevents residual costs from entering other customer bills, not merely because it protects the utility's balance sheet. Collateral, minimum bills, parent guarantees, and exit fees should be traced to the residual claimant they actually protect.

Table 26: Which protection does the instrument actually provide?

Instrument	Protects utility?	Protects ratepayers?	Protects reliability?	Main failure mode
Collateral or letter of credit	Yes	Maybe	No	Scope, callability, priority, and payer identity.
Minimum bill	Yes	Yes, if ring-fenced	No	NPV too small or wrong billing determinant.
Curtailable service	Maybe	Yes, if credited correctly	Yes	Nonperformance, no telemetry, or weak penalty.
Parent guarantee	Yes	Maybe	No	Covered obligations only; affiliate maze.
Bring-your-own resource	Maybe	Yes, if allocated	Yes	Deliverability, accreditation, and peak-hour performance.

8.3 Evidence bands before numeric haircuts

For public communication, use evidence bands rather than percentages. Percentages are useful only as screening illustrations inside this toolkit or a docket appendix. They are not valuation evidence by themselves.

Table 27: Evidence bands for settlement resources

Evidence band	Treatment
Settled cash	Cash paid, escrowed, or irrevocably reserved for the correct ledger before the public exposure is created.
Callable secured credit	LOC, escrow, secured deposit, or callable collateral with clear beneficiary, priority, expiry, and draw conditions.
Enforceable but scoped promise	Filed tariff, service agreement, parent guarantee, or take-or-pay obligation that is legally enforceable but covers only defined costs.
Unsecured affiliate or project-company promise	Promise depends on affiliate credit, project-company solvency, or post-default recovery.
Narrative-only promise	Press release, voluntary pledge, or broad assurance without enforceable terms.
Unaccredited performance claim	Flexibility, self-supply, or resource claim without telemetry, dispatch rights, accreditation, test event, and penalty.

8.4 Illustrative valuation haircuts

The following haircuts are screening illustrations only, not valuation evidence. They are non-adjudicative starting points for public analysis. They are not legal rules and should not be used as final recovery estimates. A regulator should adjust or replace them based on state law, tariff language, customer credit, collateral priority, bankruptcy risk, and whether the instrument protects the correct ledger.

Table 28: Appendix-only illustrative haircuts for settlement resources

Settlement resource	Starting credit	Why not 100 percent?
Cash paid before construction	95–100%	Cash has already settled, except for refund rights or misallocation.
Irrevocable standby letter of credit from a strong bank	85–95%	Value depends on call conditions, expiry date, governing law, and priority.
Unconditional investment-grade parent guarantee	70–90%	Strong credit still depends on scope, enforceability, and whether the parent is the legal payer.
Parent guarantee that excludes shared or regional costs	40–70%	It may protect the utility bill while leaving capacity, water, or fiscal costs outside the promise.
Affiliate guarantee	20–50%	Affiliate credit may be weaker than the brand name suggests.
Project-company promise without collateral	10–30%	A thin SPV may not survive delay, underuse, exit, or bankruptcy.
Press release, voluntary pledge, or unenforceable side assurance	0–10%	A story does not settle a bill unless it becomes an enforceable instrument.
Unmetered flexibility claim with no penalty	0–20%	Flexibility that cannot be measured or punished should not receive much capacity credit.
Dispatchable, audited, penalty-backed curtailable load	Accredited capacity value times performance probability	Count only the value that survives measurement, event triggers, penalties, and failure externality.

Table 29: Cap and floor rules for screening haircuts

Rule	Consequence
Instrument does not cover the ledger where the exposure sits.	Value equals zero in that ledger, even if the instrument is strong elsewhere.
Unsecured promise payable only after bankruptcy or litigation.	Cannot score above low credit without collateral, escrow, LOC, or enforceable parent support.
Flexibility has no telemetry, dispatch right, accreditation, or penalty.	Cannot be counted as a capacity settlement resource.
Minimum bill is not ring-fenced to prevent residual costs from entering other customer bills.	Counts as utility cash-flow protection, not automatically as ratepayer protection.
Local fiscal benefit has no clawback, reporting duty, or enforceable payment obligation.	Counts as a narrative benefit, not a settlement resource.
Collateral expires before the relevant stress horizon.	Credit is capped at the value available before expiry, not face value.

Minimum bills and exit fees should be valued as net present values, not labels. A minimum bill can be discounted using the utility's approved carrying-cost rate or another commission-approved discount rate, with a credit spread if the payer is not investment grade. An exit fee should be compared with remaining net book value under the relevant exit year, not only with a round number of months. Capacity-market exposure should be estimated from the instrument that actually creates retail exposure: IMM counterfactuals for system scale, PJM RPM or FRR obligations for load-serving exposure, and state retail riders or class allocation for household pass-through.

8.5 Haircut calibration appendix

The percentage ranges above are not a hidden model. They are a starting point that should be calibrated from instrument-specific facts. A docket version should replace each default score with evidence about credit, callability, legal priority, scope, timing, and jurisdiction match.

Table 30: Haircut calibration components

Adjustment	Evidence to inspect
Credit rating adjustment	Parent or payer rating, audited liquidity, debt covenants, downgrade triggers, and affiliate support.
Callability adjustment	Whether the utility or public entity can draw cash before default, after default, only after litigation, or only after bankruptcy.
Bankruptcy-priority adjustment	Whether the claim is secured, senior, subordinated, unsecured, capped, stayed, or subject to lender control.
Scope-of-obligation adjustment	Whether the instrument covers direct facilities only, or also shared network upgrades, capacity exposure, water debt, tax clawbacks, and exit exposure.
Jurisdiction-match adjustment	Whether the instrument protects the ledger where the exposure sits: electric, RTO, local fiscal, water, environmental, or credit.
Timing-to-cash adjustment	Whether cash is available pre-construction, during operations, after exit, after claim process, or after litigation.

Table 31: Three calibration examples

Instrument	Calibration path	Likely treatment
Investment-grade hyperscaler parent guarantee	Confirm parent is legal obligor or unconditional guarantor; verify scope reaches direct, shared, exit, and relevant capacity exposure; check call conditions and governing law.	High credit only for covered obligations; haircut any exposure outside scope.
Thin SPV promise	Inspect assets, parent support, tenant assignment rights, lease structure, secured lenders, and bankruptcy remoteness.	Low credit unless backed by cash, LOC, escrow, or enforceable parent support.
Irrevocable standby letter of credit	Verify issuing bank, amount, expiry, evergreen clause, draw conditions, beneficiary, governing law, and whether proceeds are ring-fenced for the correct ledger.	High credit if liquid, senior, callable, and matched to the exposure.

Table 32: Default stress scenarios for the Ledger Adequacy Test

Stress scenario	What to calculate	Pass/fail question
Two-year delay	Construction costs incurred before load arrives, minimum-bill start date, carrying cost, and cancellation payment	Does the customer pay for capital committed during the delay?
60 percent realized load	Fixed-cost recovery at actual usage compared with contracted demand and minimum bill	Does the minimum bill cover the underuse gap?
Exit in year 5	Remaining net book value, exit-fee NPV, collateral call, and replacement-load probability	Does the exit package cover stranded assets or only a transition period?
Credit downgrade or affiliate default	Collateral value after haircut, callability, priority, parent depth, and bankruptcy delay	Does the recovery remain available before other creditors block it?
Regional capacity price shock	LSE obligation, RPM or FRR exposure, retail rider, and class allocation	Does the customer-specific instrument capture the regional cost?
Curtailment failure	Expected flexibility value, failure penalty, telemetry record, and reliability externality	Does the penalty cover the system cost of nonperformance?
Local infrastructure underuse	Remaining water, road, or public-safety debt service after cancellation or underuse	Are taxpayers or local utility users left with stranded public assets?

Table 33: Required Stress Case Matrix

Project type	Required stress cases	Optional stress cases	Trigger for extra review
Hyperscaler owned or operated campus	Delay, underuse, exit, shared-cost scope, and regional capacity shock.	Parent downgrade or affiliate restructuring.	Parent guarantee scope unclear, large shared upgrades, or RTO forecast inclusion.
Hyperscaler lease from colocation provider	Tenant departure, landlord default, assignment change, underuse, and exit.	Low-load and parent-credit stress.	Tenant is not legal payer or landlord lacks parent support.
Speculative developer or queue holder	Cancellation, queue withdrawal, no end user, credit default, and delayed service.	Local infrastructure underuse.	No executed end-user agreement or thin SPV.
Behind-the-meter or co-located generation	Standby failure, fuel risk, emissions limits, deliverability, and grid-import shock.	Capacity accreditation loss.	Claims to avoid grid costs, capacity obligations, or standby charges.
Flexible or interruptible load	Curtailment failure, telemetry failure, event response, and penalty adequacy.	Loss of capacity accreditation.	Claims flexibility credit, lower minimum bill, or capacity offset.
Large public-infrastructure or water-dependent site	Local infrastructure underuse, water debt, road/fire staffing, and exit.	Drought-year or permit-limit stress.	Public bonds, tax incentives, or water-authority capacity reservation.

8.6 When data centers improve the public ledger

A fair test counts benefits as well as risks. Some data centers may produce high load-factor revenue, support new transmission or generation as anchor load, offer verified flexibility, reduce forecast risk through long take-or-pay contracts, or provide local tax and workforce benefits. Those benefits should enter the ledger, but with the same discipline as costs.

Supporters should win when their benefits settle

A data center should be treated as improving the public ledger when the revenue-cost surplus survives low-load stress; capacity obligation is fully allocated or offset; shared infrastructure has later-user credits; local fiscal NPV remains positive after abatements and water, road, school, and public-safety costs; flexibility is accredited and penalty-backed; and parent support covers direct, shared, and exit exposure.

Two dashboards, not one blended pot

Settlement Adequacy Dashboard: asks whether each same-ledger payment chain closes: electric ledger gap, RTO/capacity ledger gap, local fiscal ledger gap, water ledger gap, credit ledger gap, and technical flexibility ledger gap.
Public Welfare Dashboard: asks whether a decision-maker is explicitly making a broader public-welfare tradeoff: aggregate benefits, aggregate costs, winners, losers, compensation, distributional effects, and residual claimant.
A public-welfare finding may justify a policy choice. It does not by itself settle a payment obligation.

Table 34: Benefit categories: where upside claims belong

Benefit type	Examples	Where it can count
Settlement benefits	Cash payments, minimum bills, enforceable tax clawbacks, callable collateral, parent guarantees, project-specific water payments.	Settlement Adequacy Dashboard, but only for the ledger the instrument legally covers.
System benefits	Deliverable capacity, accredited flexibility, later-user credits, shared infrastructure with refund rules, grid resilience resources.	Settlement dashboard only if deliverable, accredited, and linked to the cost it offsets; otherwise public-welfare dashboard.
Strategic or public-welfare benefits	Innovation, national competitiveness, regional economic strategy, workforce ecosystem, national-security claims.	Public Welfare Dashboard only; they do not pay electric bills, water debt, or PJM capacity charges unless converted into an enforceable transfer.

Table 35: Settlement dashboard versus public-welfare dashboard

Dashboard	What it can conclude	What it cannot do
Settlement Adequacy Dashboard	Whether enforceable settlement resources cover same-ledger exposure under stress.	It cannot use jobs, tax revenue, or broad public benefits to pay a specific electric, water, or capacity obligation unless an enforceable transfer exists.
Public Welfare Dashboard	Whether an authorized public body knowingly accepts a cross-ledger trade-off after naming benefits, costs, winners, losers, compensation, and residual claimants.	It cannot pretend the payment chain has settled merely because aggregate benefits exceed aggregate costs.

Benefits and costs use the same evidentiary standard. Jobs, investment, innovation, resilience, or tax revenue should not be credited at full value if they appear only in a press release. They need legal instruments, reporting duties, clawbacks, deliverability tests, and stress-case survival just like minimum bills and collateral do.

Table 36: Benefit ledger: upside claims that must also be tested

Claimed benefit	Evidence needed	Stress test
Incremental utility margin	Cost-to-serve study showing revenues exceed incremental cost by class and by cost bucket	Does the surplus survive low-load, delayed-ramp, and high-capacity-cost scenarios?
Anchor load for new infrastructure	Transmission, generation, or substation plan showing broader public use and later-user credit rules	If the data center exits, does the asset still serve other customers or become stranded?
Capacity or resource contribution	Deliverability study, accreditation, performance obligation, and penalty-backed contract	Does the resource perform at peak hours and during system stress?
Verified flexibility value	Curtailment telemetry, event triggers, audit rights, and failure penalties	Does the flexibility reduce capacity need before scarcity costs are incurred?
Local fiscal benefit	NPV of taxes after abatements, PILOTs, fees, school-district allocation, and clawbacks	Does the local ledger remain positive after water, road, emergency, noise, and environmental costs?
Workforce or community investment	Binding agreement, dollar amount, eligibility rules, and reporting obligation	Does the commitment survive ownership change, delay, or automation of operations?
Publicly usable infrastructure	Agreement showing who owns, maintains, and can use the infrastructure after completion	Is the asset available to the public or dedicated mostly to the project?

Table 37: Supporter Evidence Protocol

Supporter claim	Required evidence discipline
Commissioned study shows surplus value	Disclose sponsor, data sources, assumptions, included and excluded cost buckets, and whether the sponsor reviewed the study before release.
Cost-to-serve surplus	Make the study reviewable by staff, public counsel, or consumer advocate under protective order; separate generation, transmission, distribution, capacity, and customer-class effects.
Surplus survives stress	Test low-load, delayed-ramp, high-capacity-cost, exit, and credit-default scenarios rather than base case only.
Local fiscal benefit	Include tax abatements, school allocation, water, roads, emergency services, public safety, administrative cost, and clawbacks.
Flexibility benefit	Require telemetry, dispatch rights, test events, accreditation, and penalty-backed performance.
Shared infrastructure benefit	Provide later-user credits, refund mechanism, beneficiary allocation, ownership, maintenance payer, and public-use rights.

What would make opponents lose fairly?

Opponents should concede a project has passed the framework when reviewable evidence shows that: cost-to-serve surplus survives 60 percent load stress; direct and shared network costs are covered by enforceable instruments; regional capacity exposure is assigned or offset; local water and school burdens are covered; the parent guarantee reaches the true legal payer and shared obligations; public counsel or staff reviewed confidential terms; and no material residual claimant remains unnamed.

Table 38: Benefit valuation worksheet

Benefit	Evidence	Credited value	Residual uncertainty
Incremental margin	Cost-to-serve study by class and bucket	Revenue surplus after low-load stress	Class allocation and capacity pass-through.
Local tax	NPV fiscal note after abatements	Net tax/PILOT value after public costs	School split, clawback, and default.
Flexibility	Telemetry, test event, dispatch rights	Avoided capacity value times performance probability	Failure externality and penalty adequacy.
Public infrastructure	Access, ownership, and maintenance agreement	Share usable by public or later users	Maintenance payer and future-use risk.

Later-user credit rule

Shared infrastructure can be fairly assigned to an early data-center project only if the record shows how later beneficiaries will repay, refund, or share the cost. A later-user mechanism should specify refund term, cap, trigger, ownership, depreciation, and the customer class credited. Without that mechanism, “future public benefit” should not offset current public risk at full value.

Table 39: How to use the Ledger Adequacy Test

Step	What to calculate	Evidence source
Define the project	Contracted MW, expected load factor, service date, ramp schedule, customer legal entity, and affiliate guarantees	Interconnection request, service agreement, utility queue report, customer filing
List project-caused costs	Direct interconnection, substations, transmission, generation/resource adequacy, shared upgrades, water, roads, public services	Rate case, IRP, certificate proceeding, RTO studies, local fiscal analysis
List settlement resources	Upfront payments, minimum bills, take-or-pay term, collateral, exit fee, resource contribution, tax clawback	Tariff, special contract, commission order, public-benefit agreement, bond or credit disclosure
Stress the forecast	Delay, cancellation, 60 percent usage, early exit, bankruptcy, regional capacity spike, tax revenue shortfall	Commission testimony, sensitivity cases, public counsel discovery, investor materials
Find the gap	Compare resources and exposures in dollars, months, and megawatts	Public ledger prepared by regulator, public counsel, journalist, or local government

Table 40: Ledger Adequacy Scorecard schema

Scorecard field	What must be entered for each cost item
Cost item and amount	The dollar amount or range, with source and confidence level.
Timing	When the cost arises and when the settlement resource becomes available.
Caused by	Whether the cost is customer-specific, shared, regional, forecast-driven, or local public infrastructure.
Legal payer	The entity legally obligated to pay, including parent, affiliate, landlord, tenant, or SPV.
Instrument	Tariff, service agreement, guarantee, letter of credit, exit fee, local agreement, RTO rule, or commission order.
Public docket	The proceeding, docket, filing, or proxy record where the public can inspect the claim.
Collateral and priority	Whether recovery is cash, deposit, letter of credit, parent guarantee, unsecured promise, or post-default claim.
Residual claimant	Who pays if the instrument fails: large-load customer, utility shareholder, other ratepayers, taxpayer, creditor, or future customer.

Table 41: Ledger Adequacy Certificate

Ledger item	Amount, timing, and payer	Instrument, haircut, and jurisdiction	Residual claimant and evidence status
Direct interconnection	\$X; pre-service; project company or parent	Service agreement; 90%; PUC	Ratepayers or shareholders; public or confidential
Shared network upgrade	\$X; construction; utility, RTO, or customer	RTO tariff, rider, or upgrade agreement; 50%; FERC/RTO/PUC	Regional customers; incomplete or contested
Capacity exposure	\$X; delivery year; LSE or customer	RPM, FRR, or customer charge; 40%; PJM/FERC/state	Retail classes; contested
Water infrastructure	\$X; bond term; water authority or project	Water agreement; 70%; local authority	Water users; public
Tax abatement	\$X; 20 years; county and project	Abatement agreement; 60%; county and school district	Taxpayers or schools; public

The certificate is deliberately disciplined, not mechanically punitive. It asks the public decision-maker to name the amount, the payer, the legal instrument, the evidence band or haircut, the jurisdiction, the residual claimant, and the evidence status. If a line cannot be filled, the correct label is not automatically fail. It may be not applicable, not triggered, confidential but reviewable, contested, or unresolved. The key rule is narrower: an unresolved line should not be used to justify public-cost recovery until the decision-maker has reviewable evidence or makes an explicit public-welfare finding.

9 Translate the Data Center Into Balance Sheets

The usual public conversation says: "AI data centers use a lot of electricity."

The Money View says something more precise: an AI data center is a payment chain. It ties together the power grid, land, water, local budgets, public credit, and capital-market expectations. That may sound abstract, but the practical question is simple: which promises are private, and which ones can end up on the public ledger?

9.1 The technology company's balance sheet

For a large cloud provider or AI company, a data center is an asset. It may support model training, AI services, cloud revenue, platform control, national security claims, and investor confidence in future growth.

But the asset has liabilities behind it: capital spending, leases, project debt, long-term power contracts, minimum bills, interconnection costs, exit fees, and possible guarantees. If a company says, "We will pay our own way," the public should ask: where is that promise written? Is it an upfront payment, a minimum bill, a guarantee, a long-term obligation, or only a press release?

9.2 The utility's balance sheet

For an electric utility, a new data center can also look like an asset. More load can mean more sales. New substations, transmission lines, and grid equipment can become utility property. If regulators approve, those investments may enter the rate base and be paid back through future customer bills.

But the utility must finance those investments before all the promised power bills arrive. It may issue debt, commit equity, take construction risk, and ask regulators to approve cost recovery. This is where the public stakes become clear: one party's asset may become another party's future bill.

In Money View language, the utility is a balance-sheet intermediary. It turns a private load request into physical infrastructure and then asks the regulatory system to turn that infrastructure into future cash flows. A public utility commission is therefore not just a referee. It is part of the settlement machinery. It decides which promises are strong enough to protect ordinary customers.

9.3 The utility incentive problem

The utility is not always a neutral pass-through institution. Under cost-of-service regulation, a utility may earn an allowed return on capital investments that enter the rate base. A larger data-center load forecast can therefore support a larger capital plan, and a larger approved capital plan can become a long stream of regulated cash flows.

That does not mean every utility proposal is opportunistic. It means the public ledger must ask two questions at once: did the data center sign the obligation that pays for the cost it caused, and did the utility have an incentive to overbuild around a speculative forecast?

Good settlement design should therefore include milestone-based approvals, used-and-useful review, shareholder risk sharing for forecast error, disallowance for speculative load, prudence triggers tied to actual customer commitments, and large-load-specific riders with true-ups. Otherwise the data center's forecast can become the utility's asset before it becomes the customer's enforceable bill.

Table 42: Utility incentive chain

Step	Money View interpretation
Large-load forecast enters planning	A future customer promise becomes the basis for present capital commitments.
Forecast justifies new capex	The utility converts expected load into substations, transmission, generation, or fuel infrastructure.
Capex enters rate base	The asset becomes a regulated claim on future customer cash flows.
Allowed return and riders speed recovery	The utility may receive a return while customers carry forecast risk through rates or riders.
Prudence review arrives after commitments	Once infrastructure is built or contracted, disallowance becomes politically and legally harder.
Customer exits or underuses service	The utility may still argue the asset was prudent or used and useful, shifting residual risk to customers unless ring-fenced.

Table 43: Forecast-error sharing mechanism

Forecast status	Recommended treatment
No executed service agreement	Forecast may support preliminary study or queue screening, but should not support major construction cost recovery without a separate public finding.
Executed agreement, no deposit or collateral	Forecast may support limited engineering; construction commitments should require milestone payments, deposit, collateral, or shareholder sharing.
Executed agreement with milestone schedule	Utility recovery should track customer milestones, ramp schedule, and actual readiness rather than headline MW alone.
Project misses milestone or delays service	Carrying costs should be charged to the customer or shared with shareholders unless the commission makes a supported finding that the asset is independently used and useful.
Speculative or affiliate-split load	Require aggregation, queue-withdrawal penalties, later-user reassignment rules, and proof that the legal payer cannot strand obligations through affiliate separation.
Asset remains unused after a defined review point	Trigger used-and-useful review, deferred recovery, customer collateral draw, shareholder sharing, or reassignment to later beneficiaries.

9.4 Households, businesses, and local government

For households, the data center may not be an asset at all. It may appear only as a monthly electric bill. Unless regulators assign costs carefully to large-load customers, some grid expansion costs, backup costs, capacity costs, and stranded asset risks can be spread across ordinary customers.

For local government, a data center may bring future tax revenue. It may also bring public commitments: tax breaks, land support, water infrastructure, roads, permitting capacity, emergency services, and political promises. The real question is not whether the project brings revenue. The question is what the net public ledger looks like after incentives and infrastructure costs.

The electric bill is only the inner ledger. The outer ledger includes public finance and local externalities. A serious review should ask for the value of property-tax abatements, water and wastewater upgrades, road and transmission-siting costs, emergency-service needs, backup diesel or gas permits, school-district revenue effects, and enforceable local hiring commitments. If those items are missing, the project may still be profitable for the company while being less profitable for the community.

Net local fiscal test

$$\begin{aligned}\text{Net local value} = {} & \text{NPV of property tax after abatements}\\ & + \text{sales/use tax, fees, PILOTs, and income effects}\\ & - \text{NPV of abatements and school-district losses}\\ & - \text{water, road, emergency, environmental, and administrative costs}\end{aligned}$$

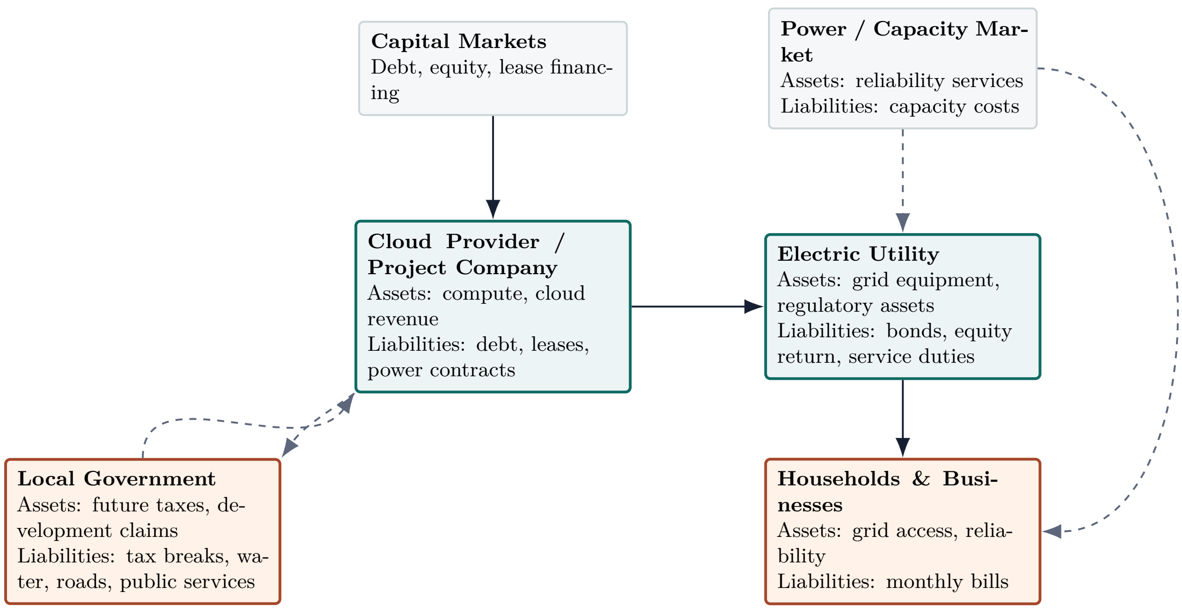
If the result is positive, the local ledger may support the project even if residents still need electric-bill protections. If it is negative, a project can be privately profitable and still rely on public balance sheets.

Table 44: The outer public ledger

Public ledger item	What to ask for	Why it matters
Tax incentives	Abatement schedule, clawback terms, school-district impact, and net present value of foregone taxes	Headline investment can hide a long public revenue concession.
Water and wastewater	Expected withdrawals, cooling technology, required public infrastructure, and who pays debt service	A low electric subsidy can coexist with a high water-infrastructure subsidy.
Backup generation and air permits	Diesel or gas capacity, operating limits, emissions controls, and emergency-operation rules	Reliability claims can shift air-quality costs to nearby residents.
Roads, land use, and public safety	Road upgrades, fire protection, noise mitigation, and emergency-response staffing	Local services can carry costs that do not appear in the power tariff.
Transmission siting	Easements, compensation, routing, and affected communities	Regional grid benefits can be concentrated into local land-use burdens.

10 Balance Sheet Graph: Put the Debate on One Page

If we listen only to the public story, an AI data center sounds like a development project. If we draw the balance sheet graph, it looks like a credit network.



Solid lines: private or quasi-private payment promises, including financing, interconnection, and bill recovery.
Dashed lines: public or system-wide promises, including tax breaks, capacity costs, and reliability costs.

Figure 1: A simplified balance sheet graph for an AI data center. The issue is not one electricity number. The issue is how payment promises move across public and private balance sheets.

The graph changes the conversation. Instead of asking only whether the project is "good for growth," we ask where the growth shows up as an asset and where it shows up as a liability.

If the company truly pays its own way, the company's balance sheet should show enforceable obligations. If costs enter the utility's rate base, the utility receives an asset and customers provide the future cash flow. At that point, a "private investment" has become partly a public credit structure.

11 Sources and Uses: Where the Money Comes From and Where It Goes

The second Money View tool is a sources-and-uses account. It is a simple project ledger.

Table 45: Sources and uses for an AI data center project

Sources of funds	Uses of funds
Cloud provider cash	GPUs, servers, networking equipment, software infrastructure
Project debt and lease financing	Land, buildings, cooling systems, security, internal power systems
Utility debt and equity	Substations, interconnection facilities, transmission upgrades, distribution upgrades
Future customer bills	Assets added to the rate base, capacity charges, reliability costs
Local tax incentives	Tax abatements, land discounts, roads, water support, permitting support
Power purchase agreements and clean-energy certificates	Annual energy matching, renewable claims, emissions accounting
Public infrastructure support	Emergency services, roads, water, wastewater, local fiscal commitments

The point is not that large investment is bad. The point is that every use of funds has a source of funds.

If the servers are paid for by the cloud provider, that is private capital spending. If a substation is built by the utility and added to the rate base, future customers support that asset through electric bills. If local government offers tax breaks, the public sector gives up future revenue. If capacity prices rise because of new large loads, the cost may be shared across many customers. If expected demand fails to show up, stranded asset risk must land somewhere: the company, utility shareholders, creditors, ratepayers, or taxpayers.

The hard lesson

A project can look like a win-win story in a press release. In the sources-and-uses account, someone supplies the money, someone uses the money, and someone holds the risk if the forecast is wrong.

12 A Pro Forma Ledger: A 500 MW Scale Translation

The previous sections name the ledger. This section starts to calculate it. The numbers below are illustrative, not an estimate for any specific project. They are placeholders for docket evidence. In a real proceeding, each input should come from an interconnection agreement, utility cost study, IRP, rate case, RTO filing, service agreement, or local fiscal analysis.

This is not a bill-impact estimate. It is a scale translation. The point is not that $1.2 billion, 11 percent, or one million households are the right numbers everywhere. The point is that the public should be shown the numbers that are right for the project in front of it. Any formal bill-impact claim must first complete a rate-class allocation model.

Table 46: Illustrative assumptions for a 500 MW data center

Assumption	Illustrative value
Contracted load	500 MW
Utility-owned grid and capacity-related assets built for the project	$1.2 billion
Upfront contribution by the data center	$300 million
Net amount entering or relying on utility cost recovery	$900 million
Annual carrying charge on recoverable assets	11 percent, including return, depreciation, taxes, and operations assumptions
Annual fixed-cost recovery need	$99 million per year
Residential customers over which residual costs could be spread if not ring-fenced	1,000,000 customers
Minimum bill if customer takes firm service	85 percent of contracted demand
Exit-fee proxy	36 months of minimum charges

Under these assumptions, if the $99 million annual fixed-cost recovery were fully socialized across one million households, it would equal about $99 per household per year, or $8.25 per month. If a tariff, contribution, or contract protects 70 percent of that fixed cost, the residual is $29.7 million per year, or about $2.48 per household per month. That is why the exact contract matters: small percentages of a large infrastructure bill can still be visible on household bills.

That household figure is only a first-pass translation. Actual bill impact depends on which cost bucket receives the charge, which customer classes share it, and which billing determinant is used. A dollar assigned through a transmission rider may not move the same way as a dollar assigned through a generation capacity charge, a distribution demand charge, or a fixed customer charge.

The household number is not the estimate

The household number is only a translation device. The allocation mechanism is the estimate. A serious bill-impact claim must show the cost bucket, rate class, billing determinant, rider, coincident-peak rule, municipal or cooperative exposure, low-income protection, and any shareholder sharing or disallowance.

Table 47: From average bill estimate to rate-class impact

Cost bucket	Allocation question	Public question
Transmission	Is the cost allocated by coincident peak, network service, zone, or retail pass-through rider?	Are residential and small-business customers exposed to a forecast-driven regional charge?
Distribution	Is the upgrade direct-assigned to the large-load customer or spread through class rates?	Does the data center pay for local facilities built mainly for its service?
Generation or resource adequacy	Is the cost recovered through capacity charges, base rates, riders, or customer-provided resources?	Does the customer pay for the system capacity reserved for it at peak hours?
Customer class allocation	How are costs split across residential, commercial, industrial, municipal, and cooperative customers?	Does the average household number hide larger burdens on particular classes or communities?
Billing determinant	Is recovery based on kWh, kW demand, customer charge, coincident peak, or contracted capacity?	Does the rate design charge the customer that drives the cost, or merely the customer class easiest to bill?
Shareholder sharing or disallowance	Do utility shareholders absorb any forecast error or imprudent investment?	Does the utility have an incentive to overbuild for speculative load?

Table 48: Minimum fields for a formal bill-impact model

Required field	Why it matters
Cost bucket	Transmission, distribution, generation, capacity, water, tax, or local infrastructure costs flow through different mechanisms.
Jurisdiction	State PUC, FERC/PJM, county, water authority, or environmental agency may control different parts of the claim.
Allocator	Costs may be allocated by kWh, kW demand, coincident peak, contracted capacity, tax base, or direct assignment.
Affected customer class	Residential, commercial, industrial, municipal, cooperative, and low-income customers may not face the same exposure.
Low-income impact	A small average increase can be large for households already near affordability limits.
Municipal or cooperative exposure	Public-power and cooperative customers can face different pass-through rules from investor-owned utility customers.
Shareholder/ratepayer split	Disallowance, sharing mechanisms, or riders determine whether forecast error lands on investors or customers.

Table 49: How forecast failure changes the ledger

Scenario	What happens in the ledger	Public question
No minimum bill, actual use reaches only 60 percent of forecast	Up to 40 percent of expected fixed-cost recovery may be exposed unless recovered from the customer by other means. On the illustrative $99 million annual cost, that is $39.6 million per year.	Does the tariff assign the shortfall to the customer, utility shareholders, or other ratepayers?
85 percent minimum bill, actual use reaches only 60 percent	The minimum bill reduces the shortfall from 40 percent to 15 percent of expected fixed-cost recovery. On the illustrative ledger, the remaining exposure is $14.85 million per year.	Is 85 percent high enough for assets built almost entirely for one customer?
Project exits after year 5; asset life is 30 years	Straight-line depreciation would leave roughly $750 million of net book value on a $900 million asset. A 36-month exit-fee proxy based on 85 percent of annual fixed-cost recovery would be about $252 million.	Does the exit fee cover the stranded asset, or only buy time while the utility searches for replacement load?
Parent guarantee weakens or project company files bankruptcy	The payment promise may fall below secured lenders, landlords, equipment lessors, or other claims.	Is the guarantee unconditional, senior, liquid, and enforceable against the parent, or only against a thin project company?

The simple stress test changes the public debate. A minimum bill is not automatically enough. An exit fee is not automatically enough. A parent guarantee is not automatically enough. The question is whether each device covers the specific cash-flow gap it is supposed to cover.

Table 50: Three illustrative Ledger Adequacy scenarios

Scenario	Settlement resources	Downside exposure	What the test would likely show
Low-risk case	High upfront contribution; long take-or-pay term; unconditional investment-grade parent guarantee; tested interruptible service; enforceable local clawbacks	Mostly direct facilities, limited shared-network exposure, reduced peak-capacity risk, transparent fiscal costs	The project may pass if fixed payments and guarantees cover remaining net book value under delay, underuse, and exit.
Middle case	AEP Ohio or Virginia-style protections: minimum bill, collateral, exit fee, separate large-load class, and some upfront payments	Meaningful recoverable grid assets, some regional capacity exposure, and uncertain local fiscal costs	The project may pass direct-cost review but still need proof on regional capacity spillovers, exit-fee sufficiency, and local public costs.
High-risk case	Low upfront payment; short commitment; thin SPV; weak collateral; vague clean-energy claim; no local clawback	Shared network upgrades, forecast-error risk, standby service, regional capacity cost, and public infrastructure costs may remain outside the customer contract	The likely ledger gap is negative; recommended treatment is unresolved cost-shift risk until the applicant produces adequate evidence.

The middle case is where most public fights will sit. It will not be enough to say "there is a minimum bill." The test asks whether the minimum bill, collateral, exit fee, and guarantee cover the specific downside case: project delay, 60 percent usage, early exit, credit deterioration, or regional price shock.

12.1 A PJM back-of-the-envelope check

PJM reported that forecasted peak load for the 2026/2027 delivery year rose by more than 5,400 MW year over year, driven largely by data center expansion, electrification, and economic growth. Monitoring Analytics estimated that actual and forecast data-center load increased combined capacity-market revenues for the 2025/2026 and 2026/2027 auctions by about $16.6 billion. Its later 2027/2028 analysis extended the market-level counterfactual: for the 2025/2026, 2026/2027, and 2027/2028 base residual auctions together, it estimated about $23.1 billion in increased costs to other PJM customers associated with existing and forecast data-center load.[1]

Those figures should not be converted into a pro rata bill for one project. Capacity markets are nonlinear. The clearing price depends on the Variable Resource Requirement curve, zonal constraints, supply offers, capacity accreditation, import limits, mitigation rules, and the timing of the forecast. Monitoring Analytics itself reports counterfactual auction results, not a simple dollar-per-MW allocation rule. For the 2026/2027 auction alone, it estimated that including existing and forecast data-center load increased RPM revenues by about $7.27 billion, or 82.1 percent, under one reported scenario.

The right lesson is narrower but stronger. A 500 MW project cannot be assigned a mechanical share of the PJM-wide revenue increase. But the counterfactual shows why a project's impact may be larger than its direct utility bill: forecast load can enter the market-clearing mechanism before households ever see a line item. The Money View question is therefore not "what is this project's exact pro rata share?" It is "which legal instrument keeps uncertain forecast-driven capacity costs from being spread to customers who did not cause them?"

[1]Sources: PJM 2026/2027 capacity auction release and Monitoring Analytics analyses of the 2026/2027 and 2027/2028 RPM Base Residual Auctions, listed in References.

Table 51: PJM settlement chain for forecast-driven capacity costs

Settlement step	What happens	Where assignment can break down
Data center load forecast	Utility, customer, and regional planners translate expected projects into future load.	Forecast uncertainty enters before all projects are built or fully contracted.
PJM peak forecast	Regional peak expectations feed the reliability requirement.	A local load can affect a regional requirement.
RPM clearing or FRR planning	Capacity is procured through auction or fixed resource requirement planning.	Market clearing is nonlinear; there is no simple dollar-per-MW bill.
LSE capacity obligation	Load-serving entities absorb capacity costs under PJM rules.	The LSE may not be able to isolate costs by the specific large-load customer.
State retail pass-through	State tariffs, riders, or rate cases determine retail recovery.	State tools may assign local grid cost but miss regional market spillovers.
Customer class billing	Costs are allocated across classes and billing determinants.	Households may see charges without a line item labeled "data center."

Status as of May 22, 2026, 19:14 EDT; update PJM and FERC rule status before filing or publication.

Some links in this chain require state tariff design. Others require RTO or FERC rule changes. That is why a state-level large-load tariff can be necessary and still incomplete.

12.2 Regional capacity causation module

Market-level evidence proves that a regional pooling pathway exists; it does not by itself prove a project-level bill. A defensible assignment of regional capacity exposure should move through four causation layers.

Table 52: From PJM market evidence to project-level responsibility

Causation layer	Evidence standard	What it can decide
Forecast inclusion causation	Strong: executed service agreement, deposit, milestone schedule, utility/LSE forecast entry. Medium: advanced interconnection study. Proxy: queue position or public announcement. Unknown: speculative or duplicative request.	Whether the project should count in load forecasts used for resource procurement.
Peak contribution causation	Strong: coincident-peak profile, contracted demand, telemetry, and curtailment rights. Medium: modeled load shape. Proxy: generic data-center load factor. Unknown: no profile or no curtailment data.	Whether the load adds to peak resource need or can be discounted for verified flexibility.
Market-clearing causation	Strong: RTO counterfactual or zonal sensitivity showing effect on VRR curve, constraint, FRR need, or clearing price. Medium: LSE capacity obligation analysis. Proxy: systemwide IMM result. Unknown: no market sensitivity.	Whether the load is linked to market-level cost movement rather than only local facilities.
Retail pass-through causation	Strong: rider, rate case, LSE bill, class allocator, and customer-specific charge. Medium: proposed tariff. Proxy: historical pass-through pattern. Unknown: no retail allocation record.	Which retail classes or customers become residual claimants if the cost is not assigned to the large load.

Regional capacity decision rule

If only market-level evidence exists, the finding should be spillover risk, not project-level dollar liability. Project-level assignment requires forecast evidence, peak contribution evidence, market-clearing or obligation evidence, and a retail pass-through pathway. Where project-level assignment is impossible, regulators can still require qualifying capacity, bilateral capacity contracts, curtailability, regional adders, or disclosure before public-risk shifting.

Table 53: Current assignment versus proposed ledger remedy

Exposure	Current assignment mechanism	Gap under current rules	Proposed ledger remedy
RPM capacity price impact	LSE obligation and state retail pass-through.	Project-specific isolation may be impossible without additional tariff or RTO rule.	Bilateral capacity obligation, qualifying capacity, regional adder, or large-load-specific retail assignment where authorized.
Reliability Backstop Procurement charge	EDC/LSE allocation if adopted, with state allocation rules still evolving.	State assignment uncertain; residual claimant may be retail classes.	Large-load-specific rider, bilateral transfer, or capacity obligation tied to forecast-driving load.
Connect-and-manage curtailment value	Program-specific accreditation or emergency operating protocol.	No ordinary capacity credit without operator control and measurement.	Telemetry, dispatch rights, test event, accreditation, and failure penalty.
Transmission upgrade spillover	FERC/RTO allocation plus state pass-through review.	Local tariff may not capture regional transmission cost.	Later-user credits, beneficiary analysis, direct assignment where cost is customer-specific, or explicit welfare finding for socialized upgrades.

13 Why Annual Electricity Use Is Not Enough

The International Energy Agency estimates that data centers used about 415 TWh of electricity worldwide in 2024, roughly 1.5 percent of global electricity use, and could use about 945 TWh by 2030. Lawrence Berkeley National Laboratory estimates that U.S. data centers used about 176 TWh in 2023, or 4.4 percent of total U.S. electricity use, with 2028 scenarios ranging from 325 to 580 TWh.[2]

These numbers matter. But they are not the whole issue.

The electric system is stressed not only by annual energy demand, but by local bottlenecks and peak-hour needs. A grid must deliver power at a specific place, at a specific hour, while maintaining reliability. That requires capacity, transmission, backup resources, and operating reserves.

This is why a renewable power purchase agreement does not automatically settle the public debate. A power purchase agreement can match annual energy on paper. It does not necessarily solve the physical problem of hourly delivery at a constrained location.

[2]Sources: International Energy Agency, *Energy and AI*; Lawrence Berkeley National Laboratory, *2024 United States Data Center Energy Usage Report*, listed in References.

A plain-English translation

A clean-energy contract may answer the question, "How much renewable energy did the company buy over the year?" It may not answer, "Who supplies reliable power to this data center on a hot evening when the local grid is constrained?"

14 Two Grid Models, Two Cost-Shifting Paths

Some data center costs appear locally, in substations or distribution upgrades. Others enter regional markets or state-regulated planning systems. The United States does not have one grid-cost model. It has at least two pathways that matter for public debate.

14.1 Market-pooling regions: PJM and capacity-market spillovers

In PJM, the regional grid operator serving all or parts of 13 states and the District of Columbia, the capacity market is designed to secure enough future power supply to meet forecast demand.

Capacity is not the same thing as energy. Energy is the electricity used over time. Capacity is the promise that enough resources will be available when the system is under stress. In everyday terms, the capacity market is partly an insurance market for reliability.

That design creates a pool. If forecast demand rises sharply, the capacity price can rise for many customers, not only the new large load that helped drive the forecast. The cost can move before a planned data center is fully built, because the market prices the need to be ready for future peak demand.

PJM has said that its forecasted peak load for the 2026/2027 delivery year increased by more than 5,400 MW year over year, driven largely by data center expansion, electrification, and economic growth. Monitoring Analytics, the independent market monitor for PJM, went further: it concluded that data center load growth was the primary reason for recent and expected capacity market conditions, including tight supply-demand balance and high prices.

The market evidence now comes in two layers. The historical two-auction evidence is already large: Monitoring Analytics estimated that actual and forecast data-center load increased combined capacity-market revenues for the 2025/2026 and 2026/2027 auctions by about $16.6 billion. The updated three-auction evidence is larger still: in its 2027/2028 analysis, Monitoring Analytics estimated about $23.1 billion in increased costs to other PJM customers across the 2025/2026, 2026/2027, and 2027/2028 base residual auctions associated with existing and forecast data-center load.[3]

These numbers are not project-level bills. They are market-level counterfactuals. Their importance is that forecast-driven data-center load can create settlement costs outside a retail tariff, before an ordinary customer sees a line item and before a state commission can easily assign those costs to one named project.

Market-level causation does not automatically create project-level legal responsibility. A single 500 MW load may have little effect in one auction state and a large effect in another if it moves the market across a clearing-price threshold. The governance rule is narrower: market-level counterfactuals prove spillover risk; project-level assignment requires an enforceable RTO, FERC, state tariff, bilateral capacity contract, qualifying capacity contribution, curtailability rule, or regional cost adder.

How to read PJM evidence

The PJM evidence should not be read as a project-level bill allocator. It should be read as proof of a market-pooling pathway. Once forecast load enters a regional capacity mechanism, cost can move before the project is fully built and before a state tariff can easily assign it to one customer. The policy question is whether large-load tariffs, RTO rules, bilateral capacity contracts, and state pass-through mechanisms identify the same residual claimant.

[3]Source: Monitoring Analytics, analyses of the 2026/2027 and 2027/2028 RPM Base Residual Auctions, listed in References.

Do not pro-rate PJM counterfactuals

Do not divide PJM-wide counterfactual costs by MW and call the result a project bill. Do not cite market-level counterfactuals as legal liability. Use them to identify a pooled-cost pathway, then ask what instrument assigns or protects the residual claimant.

Table 54: How the same PJM counterfactual should be used

Permitted conclusion	What the evidence supports	What still must be shown
Market spillover risk found	Forecast data-center load can change regional capacity-market outcomes and create pooled costs outside a retail tariff.	Which forecast, zone, LSE/EDC, or customer class is exposed under current rules.
Project-level liability not assigned	A market-level counterfactual is not a dollar bill for one project and should not be divided by MW.	Whether an RTO rule, state tariff, bilateral capacity contract, or qualifying capacity contribution assigns responsibility.
State/RTO remedy needed	Existing rules may identify reliability need before they identify the final payer.	Which institution can create the assignment, ring-fence, rider, bilateral path, curtailment accreditation, or public-welfare finding.

This is where the public ledger becomes real. Even if a data center pays its own electric bill, its expected load can still affect the regional capacity pool. The question is whether that added cost is assigned to the customer that caused it or spread more broadly across the system.

Why the PJM case matters

The capacity market is a settlement mechanism. It translates future reliability needs into present payments. If a data center raises the forecast peak load but the resulting capacity cost is spread across a region, then the balance sheet graph has a hidden public leg. PJM is the market-pooling version of the risk.

14.2 May 2026 update: PJM backstop procurement as a live ledger test

The PJM story is no longer only about past capacity auctions. In 2026, PJM began advancing a package of large-load reforms: Reliability Backstop Procurement, an Expedited Interconnection Track, connect-and-manage rules for large loads that do not bring their own generation, and load-forecasting reforms that ask for more information about contractual arrangements, ramp rates, utilization factors, and duplicative requests.

PJM's own January 2026 materials describe the package as a response to integrating large loads reliably. Its February materials describe a connect-and-manage process in which certain large loads without offsetting generation could be curtailed before pre-emergency demand response. Its April Reliability Backstop Procurement design materials show the settlement problem plainly: costs may be allocated to electric distribution companies pro rata based on target MW, with bilateral transfers available to move obligations toward where large-load growth actually occurs. PJM also stated in an earlier working paper that it does not have the ability to allocate costs directly to specific loads, leaving allocation beyond the zone or EDC to states.

On May 20, 2026, Utility Dive reported that PJM planned to move a backstop reliability auction to September 2026 rather than March 2027, and that PJM urged states to develop rules to shield other ratepayers from data-center-driven costs. That is the thesis of this brief in live form: a market operator can identify the reliability need, but the final payer may depend on state cost-allocation rules, bilateral contracts, and retail pass-through design.

Table 55: PJM status check before filing or citation

Item	Status as of May 22, 2026, 19:14 EDT	Source to check	Risk if changed
Reliability Backstop Procurement design	Proposed and evolving; PJM board reportedly accelerated the auction target to September 2026.	PJM/FERC filing, PJM board letter, Utility Dive May 20 report.	Procurement target, counterparty, and cost allocation may change.
Connect-and-manage	Stakeholder-development process; curtailment framework not fully settled.	PJM CAMSTF materials and any FERC-approved tariff revisions.	Curtailment value, priority, and accreditation may change.
Bilateral-transfer or bilateral-contract path	Proposed/evolving as a way to move obligations toward large loads and suppliers.	PJM RBP design materials and stakeholder filings.	Payer pathway may change if bilateral phase is narrowed, delayed, or replaced.
State allocation rules	State-by-state and unsettled.	State PUC dockets and PJM/FERC filings.	Residual claimant may shift to LSEs, EDCs, retail classes, or large-load customers.
Emergency curtailment of large loads with backup generation	Short-term emergency authority reported in May 2026, not a permanent capacity-accreditation rule.	DOE emergency order and PJM operating notices.	Operational emergency relief may be mistaken for ordinary capacity settlement.

Table 56: Live test: PJM Reliability Backstop Procurement

Ledger question	Why it matters
Which large loads are driving the procurement target?	The public cannot assign cost without knowing whether the procurement is caused by committed projects, speculative requests, or broad load growth.
Who can satisfy the obligation through bilateral contracting?	A bilateral path may keep costs near the customer that needs the capacity, but only if the buyer, seller, term, credit, and deliverability are clear.
If a bilateral contract is not signed, who receives the RBP charge?	Costs can move to EDCs, LSEs, retail riders, or customer classes unless state rules assign them to the large load.
How are canceled or delayed loads treated?	If the forecasted load disappears after capacity is procured, the backstop resource can become a stranded market commitment.
What does connect-and-manage curtailment mean in settlement terms?	Curtailment is valuable only if the trigger, priority, telemetry, compensation, and failure penalty are enforceable.
Can state PUCs see the contracts needed to allocate costs fairly?	Without protective-order access to customer contracts and credit support, states may be asked to allocate costs with an incomplete ledger.

> Status as of May 22, 2026, 19:14 EDT; PJM Reliability Backstop Procurement, connect-and-manage, bilateral-transfer design, and state allocation rules remain evolving. Update before filing, publication, or citation.

The backstop procurement should therefore be treated as a real-time Ledger Adequacy Test. If new resources are procured for large-load growth, the public question is not simply whether PJM found supply. It is whether the RBP charge, bilateral contract, state tariff, customer guarantee, and curtailment rule identify the same residual claimant.

14.3 Vertically integrated regions: the rate-base expansion path

In much of the Southeast and parts of the West, the cost-shifting path can look different. The utility may be vertically integrated, meaning it plans generation, transmission, distribution, and retail service inside one regulated company rather than relying mainly on a forward capacity auction.

In that model, the key document is often not a capacity-market result. It is the integrated resource plan, or IRP. The utility files a long-term load forecast, argues that new demand requires new generation or transmission, asks the public utility commission to approve a plan, and later seeks to recover the cost through base rates, riders, or tracking mechanisms.

For a data center boom, the rate-base path can look like this:

1. The utility adds large-load growth to its forecast.
2. The IRP says the system needs new generation, transmission, substations, or fuel infrastructure.
3. The utility asks the commission for approval to build or contract for those resources.
4. If regulators find the investment prudent and used to serve customers, the asset can enter the rate base.
5. Customers then repay the utility over time through regulated rates, sometimes through rate-adjustment riders between full rate cases.

The public risk is not a capacity auction price spike. It is a heavy asset built under a forecast. If the forecast proves too high, or if a large customer leaves before the asset is fully recovered, the survival constraint can move to ordinary ratepayers unless the commission ring-fences the cost with minimum bills, exit fees, collateral, customer contributions, or disallowance of imprudent spending.

Two paths, same ledger question

In PJM, the risk may appear as a higher capacity-market charge. In a vertically integrated state, the risk may appear as a larger rate base, rider, or future rate case. In both systems, the public question is the same: did the customer that caused the cost also sign the obligation that pays for it?

15 Cost Causation Is Harder Than a Slogan

"The cost-causer should pay" is the right starting point, but it is not enough. The grid is not a private driveway. It is a network with direct costs, shared costs, reliability costs, future optionality, and benefits that can change over time. A substation may mostly serve one data center. A transmission upgrade may also improve reliability for other users. A new generator may be justified partly by data centers, partly by retirements, and partly by ordinary load growth.

So the public ledger needs classification rules, not just a slogan.

Table 57: A working cost-causation checklist

Cost category	Default assignment question	Possible settlement rule
Direct interconnection facilities	Would this facility be built but for this customer?	Direct assignment, contribution in aid of construction, upfront payment, or customer-specific charge.
Shared network upgrades	Which customers benefit now, and which may benefit later?	Initial assignment to the triggering customer with later credits, refunds, or reassignment if other users use the asset.
Resource adequacy and reserve margin	Does the load increase coincident peak or capacity obligations?	Capacity charge, minimum demand charge, interruptible-service credit, or customer-provided qualifying capacity.
Transmission and regional market costs	Does the cost flow through an RTO, federal tariff, or state retail pass-through?	RTO cost allocation plus retail class allocation review; state tariff alone may not capture all spillovers.
Forecast error and stranded assets	Who should bear the error if the load never arrives or leaves early?	Exit fee, collateral, shareholder sharing, disallowance of imprudent investment, or reassignment to new load.
Future beneficiary credit	If the asset later serves other customers, should the first customer receive credit?	Refund mechanism, assignment market, capacity transfer, or declining minimum bill.
Local fiscal and environmental costs	Does the electric tariff miss costs borne by the county, school district, water system, or neighbors?	Separate public-benefit agreement, tax clawback, water fee, emissions condition, or local mitigation fund.

This is where Money View can do more than rename regulation. It can keep the categories from drifting apart. The data center may pay the direct interconnection bill but leave regional capacity costs in the market pool. It may pay the utility but receive a tax abatement from the county. It may post collateral for electric service but leave water infrastructure to a local authority. A complete ledger keeps all of those claims visible at once.

15.1 When public cost sharing is legitimate

Not every shared cost is a hidden subsidy. The electric grid is a network, and some infrastructure is legitimately socialized because it serves multiple users, improves reliability, or advances a public purpose. The ledger test should distinguish hidden cost shifting from justified public cost sharing.

Table 58: Legitimate public cost sharing

Condition	What the public record should show
Broad beneficiary analysis	The asset is used by multiple customers or improves system reliability beyond the data center.
Later-user credit or refund mechanism	First movers are reimbursed or credited when later users benefit from the shared asset.
Explicit public-welfare finding	The decision-maker names winners, losers, distributional effects, compensation, and residual claimant.
No private exclusivity over public-funded asset	The data center does not receive exclusive use of assets funded by general customers or taxpayers.
Forecast risk is allocated openly	If demand fails, the record states whether the customer, utility shareholder, ratepayers, taxpayers, or later users absorb the cost.
Distributional harms are mitigated	Low-income customers, schools, water users, renters, and nearby residents receive targeted protections where they bear material burdens.

15.2 Jurisdiction map: ask the right institution

The ledger is fragmented because jurisdiction is fragmented. A state public utility commission may see the retail tariff, but not the county tax abatement. A zoning board may see land-use impacts, but not PJM capacity allocation. A local water authority may issue debt for infrastructure that never appears in an electric rate case. Public accountability requires matching each cost item to the institution that can actually see or change it.

Table 59: Where to look for each part of the ledger

Cost or risk item	Main institution	Document to inspect	Failure mode
Distribution and direct interconnection	State PUC, utility, cooperative board, or municipal utility	Large-load tariff, interconnection agreement, service agreement, cost-of-service study	Direct costs are under-assigned or recovered from broader classes.
Transmission and regional market costs	FERC, RTO/ISO, state PUC pass-through review	RTO tariff, transmission-cost allocation filing, capacity-market result, retail rider	Regional costs are pooled even when local tariff looks protected.
Resource adequacy and generation planning	State PUC, IRP process, RTO/ISO capacity market	IRP, certificate of need, capacity obligation, generation rider, bring-your-own-resource contract	Forecast load justifies new assets that later become stranded.
Tax abatements and economic development incentives	County, city, state economic-development agency, school district	Abatement agreement, fiscal note, PILOT agreement, clawback clause	Public revenue is surrendered while benefits remain headline-only.
Water and wastewater	Local utility, water authority, state environmental or public works agency	Water supply agreement, infrastructure bond disclosure, wastewater permit	Local debt service hides outside the electric tariff.
Backup generation and emissions	State environmental regulator, local air-quality agency, EPA where applicable	Air permit, emergency-generation limits, emissions-control conditions	Reliability is bought with local air-quality risk.
Land use, noise, roads, and public safety	Zoning board, county commission, transportation agency, fire marshal	Zoning approval, site plan, road improvement agreement, emergency-response plan	Local services absorb costs that never enter utility proceedings.
Credit support and default recovery	PUC, utility contract, lenders, bankruptcy court if default occurs	Parent guarantee, letter of credit, deposit agreement, security interest, bankruptcy-remoteness terms	The named customer cannot pay when the obligation comes due.

The practical instruction is simple: do not let one clean document substitute for the whole ledger. A project may pass the PUC tariff test and fail the county fiscal test. It may pass the county jobs test and fail the RTO capacity test. The public ledger is complete only when the pieces are read together.

Cross-Jurisdiction Ledger Statement

No single institution sees the full ledger. A major AI data-center project should therefore file a Cross-Jurisdiction Ledger Statement: a public-facing summary that links the PUC tariff, RTO or FERC exposure, local fiscal incentives, water and wastewater obligations, environmental permits, and credit-support documents.

Table 60: Institutional design for a complete ledger

Institutional rule	Purpose
PUC dockets should summarize local fiscal incentives and water obligations.	Electric regulators need to know whether a project that pays electric costs is receiving offsetting public support elsewhere.
County economic-development agreements should disclose the status of electric and capacity cost allocation.	Local officials should not approve tax deals without knowing whether households may later face power-system costs.
RTO large-load forecasts should count only meaningful and verifiable commitments.	Forecasts should distinguish signed service obligations from speculative or duplicative queue positions.
Consumer advocates or public counsel should receive redacted contracts under protective order.	The public needs an institutional representative who can see what ordinary residents cannot.
Each approval should identify the residual claimant.	If a cost-allocation tool fails, the record should say whether the payer is the customer, utility shareholder, ratepayer, taxpayer, creditor, or future user.

16 What Should Be in Writing

The phrase "the company will pay" is not enough. For the public, the key question is whether the promise has been converted into an enforceable obligation.

Table 61: Public questions that should be answered before costs are socialized

Question	Why it matters
Is there a large-load tariff?	It can separate data center costs from ordinary household and small-business bills.
Is there a minimum bill?	It protects other customers if the data center uses less power than forecast.
Is there an upfront contribution?	It reduces the risk that grid upgrades are financed first by the utility and recovered later from everyone.
Is there collateral or a guarantee?	It gives the utility and customers protection if the project is delayed or canceled.
Is there an exit fee?	It helps prevent stranded grid assets if the customer leaves.
Are tax incentives disclosed?	It lets residents see the net fiscal deal, not just the headline investment number.

These are not anti-technology questions. They are public-accountability questions.

16.1 The White House pledge: promise or settlement?

In March 2026, the White House announced a Ratepayer Protection Pledge. The proclamation says leading hyperscalers and AI companies will build, bring, or buy the new generation and electricity needed for data centers; pay for new delivery infrastructure upgrades; negotiate separate rate structures with utilities and state governments; and pay whether they use the electricity or not. It also says seven leading technology companies accepted the pledge.[4]

That pledge matters because it adopts the central public claim of this brief: households should not pay for data-center-driven infrastructure. But in Money View terms, the pledge is the beginning of the inquiry, not the end. A national political commitment must still be translated into state tariffs, RTO rules, service agreements, collateral, minimum bills, and local fiscal terms.

The pledge is useful because it expresses the public norm. It is not analytically necessary to the framework. The same ledger test applies if the pledge disappears, changes administrations, or is replaced by state legislation. The point is not the political label. The point is whether a public promise becomes a legal settlement instrument.

[4]Source: The White House, *Ratepayer Protection Pledge* and related proclamation, listed in References.

Table 62: Pledge-to-instrument conversion table

Pledge promise	Required legal instrument	Evidence needed	Failure if absent
Build, bring, or buy power	PPA, generation contract, deliverability study, and capacity accreditation	MW, commercial operation date, deliverability, peak performance, and penalties	Annual matching can be mistaken for peak reliability.
Pay for delivery infrastructure	Interconnection agreement, network-upgrade assignment, and transmission cost-allocation order	Direct vs. shared costs, refund rights, and later-user credit rules	The project may pay for its substation while regional costs remain pooled.
Pay whether used or not	Minimum bill, take-or-pay electric service agreement, and collateral package	Term, contracted MW, billing determinant, exit fee, and legal payer	Underuse or delayed ramp can shift fixed costs to others.
Support grid resilience	Curtailment protocol, telemetry, dispatch rights, and performance penalty	Event trigger, notice time, measurement, compensation, and failure penalty	Backup or flexibility claims may not be enforceable when the grid is stressed.
Benefit local communities	PILOT, tax clawback, water agreement, road agreement, and workforce commitment	NPV, public cost, clawback trigger, and reporting obligation	Headline jobs or investment can hide fiscal loss.

The pledge should therefore be welcomed as a public benchmark but tested as a legal and financial instrument. Its largest weakness is not the content of the promise. It is the execution layer. Electric rates and cost allocation are implemented through state PUC orders, RTO or FERC rules, filed tariffs, customer contracts, collateral postings, and local fiscal agreements. If the pledge is not embedded in those instruments, it remains a promise above the ledger rather than a settlement inside the ledger.

> The pledge is politically useful but legally incomplete. It should be treated as a disclosure trigger, not a settlement instrument.

16.2 A balancing matrix: price the risk, do not ban the project

The goal is not to scare away every large load. The goal is to match risk with price. A rigid tariff can become a defensive wall. A smarter tariff works like a risk-pricing schedule: the customer can lower its minimum bill or collateral burden by offering real value back to the grid.

Table 63: Tiered risk pricing for large data center loads

Customer commitment	Regulatory response	Why it balances growth and protection
Firm 24/7 service with no curtailment commitment	High minimum bill, full interconnection cost responsibility, and strong collateral	The customer is asking the system to reserve capacity at all times, so the customer should carry more fixed-cost risk.
Tested non-firm or interruptible service during defined grid-stress hours	Lower minimum-bill percentage or capacity charge for the interruptible portion	If the data center can reliably give capacity back to households during peak stress, it is causing less capacity risk.
Onsite generation, storage, or bring-your-own-resource plan with deliverability and performance penalties	Lower upstream capacity adder or standby charge, but not automatic exemption from grid costs	The resource helps only if it performs when the grid needs it and does not create new reliability costs.
Investment-grade parent guarantee from a hyperscaler	Lower cash collateral if the guarantee is unconditional and enforceable	A strong corporate guarantee can sit high in the hierarchy of promises without trapping unnecessary cash.
Special-purpose or speculative developer with weak credit	Cash deposit, letter of credit, escrow, or larger exit-fee security	If the project is easier to abandon, the public needs stronger protection before infrastructure is built.

This is the practical compromise. A data center that behaves like a rigid, always-on load should pay for that rigidity. A data center that can be interrupted, bring dependable resources, or post strong credit support can earn more flexible treatment. Regulation becomes a conversion table between technical flexibility and financial obligation.

16.3 Technical performance ledger

AI data centers are not merely large annual energy users. They can be concentrated electronic loads with operational characteristics that matter for reliability: ramp rates, voltage behavior, ride-through performance, power quality, telemetry, backup generation, and curtailment performance. A flexibility promise should therefore be written as a technical settlement obligation, not as a public-relations label.

This section operationalizes the verified-flexibility line in the Ledger Adequacy Test. Flexibility should reduce the customer's financial obligation only when it can be measured in real time, called by the grid operator, audited after the event, and penalized if it fails. Otherwise the customer receives credit for a capacity benefit that other customers may have to supply at settlement.

Accreditation boundary

Flexibility credit should match the market or planning product it offsets. A customer's curtailment promise should not reduce capacity, transmission, or fixed-cost obligations unless the relevant operator or commission can call it, measure it, accredit it, and penalize failure.

Table 64: Workload-to-flexibility matrix

Workload	Likely flexibility	Evidence needed before credit is granted
Model training	Potentially shiftable, but only if checkpointing, cluster scheduling, deadlines, and interconnect constraints allow interruption.	Workload scheduler evidence, tested curtailment event, recovery time, and service-level commitments.
Batch inference	Some shiftability if latency tolerance and queue management exist.	Latency class, queue-management rules, customer commitments, and measurement protocol.
Real-time inference	Lower flexibility because latency and availability are often part of the product.	SLA terms, redundancy plan, affected region, and penalty for failed curtailment.
Cloud colocation	Tenant-dependent flexibility; the landlord may not control tenant workloads.	Tenant contracts, aggregation rights, curtailment authority, and assignment of penalties.
Dedicated accelerator campus	High power density and possible ramp or power-quality risk.	Ramp controls, power-quality mitigation, cooling assumptions, telemetry, and operator testing.

Table 65: Technical performance items that should enter the ledger

Performance item	Required data or rule	Payment question
Interruptible or non-firm service	Event trigger, notice time, maximum annual hours, measurement protocol, and failure penalty	If the customer fails to curtail, does the penalty cover the capacity and reliability cost it imposed?
Ramp-rate controls	Maximum ramp up/down, coordination with grid operator, telemetry, and operational limits	Who pays for balancing or reserve costs caused by rapid load changes?
Ride-through and voltage performance	Interconnection performance standard, disturbance ride-through obligation, and testing record	Who pays for mitigation if the facility trips or contributes to voltage disturbance?
Power quality mitigation	Harmonics, flicker, reactive power, filtering equipment, and monitoring requirements	Are mitigation costs assigned to the data center or socialized through grid upgrades?
Behind-the-meter generation or storage	Deliverability, fuel availability, emissions permits, dispatch rules, and performance penalties	Does the resource actually reduce system need at peak, or only support the facility privately?
Metering and telemetry	Real-time load, curtailment verification, onsite generation output, and audit access	Can the regulator verify that the customer delivered the flexibility it was paid for?

Table 66: Flexibility accreditation template

Contract field	Required specification	Settlement question
Telemetry interval	Real-time, 1-minute, 5-minute, or other interval; recipient is utility, RTO, aggregator, or PUC staff.	Can performance be measured at the same granularity as the claimed system benefit?
Curtailment notice time	Emergency, 10-minute, 30-minute, day-ahead, or seasonal call.	Does the operator receive the response before scarcity costs are incurred?
Dispatch authority	Operator-dispatchable, utility-dispatchable, aggregator-dispatched, or customer-declared.	Is the flexibility under system control or merely a voluntary promise?
Annual and event limits	Maximum annual hours, maximum events, minimum restoration time, and recovery constraints.	Is the value durable enough for the capacity or planning product it offsets?
Failure penalty	Formula based on avoided capacity value, scarcity price, replacement procurement cost, reliability externality, and administrative cost.	Does the penalty make the public ledger whole if curtailment fails?
Workload feasibility	For training: checkpoint and restart cost. For colocation: tenant contract rights. For BTM generation: fuel, emissions, and ride-through evidence.	Does the customer actually control the load or resource it claims to offer?
Accrediting authority	RTO, utility IRP, PUC tariff, demand-response program, or local reliability protocol.	Does the credited flexibility match the market or planning product whose cost it offsets?

The ledger point is straightforward: flexibility has value only if it is measurable, deliverable, accredited in the relevant product, and backed by penalties. Otherwise it is another promise sitting below settlement.

17 Two Cases Where the Ledger Became Real

The evidence ladder is not theoretical. Regulators are already writing data-center-specific rules.

17.1 AEP Ohio: minimum demand, collateral, and exit risk

In July 2025, the Public Utilities Commission of Ohio adopted AEP Ohio's Data Center Tariff settlement. AEP Ohio says the tariff became effective on July 23, 2025. It applies to new data centers or expansions of 25 MW or greater in its service territory.[5]

The tariff shows what "show me the contract" looks like in practice. The following are not slogans. They are ledger devices.

[5]Source: AEP Ohio, *Data Center Tariff*, listed in References.

Table 67: AEP Ohio tariff provisions as Money View settlement devices

Risk	Tariff device	Money View meaning
Speculative grid queue	Load-study fee from $10,000 to $100,000	The customer must put cash behind the request before occupying planning capacity.
Canceled or delayed project	Reimbursement of 100 percent of buildout costs if the project is canceled or delayed more than 12 months before target energization	Construction risk is pushed back toward the customer instead of being left with the utility system.
Slow or uncertain ramp-up	Contract term equal to the ramp period, up to four years, plus eight years; minimum demand rules tied to contract capacity	A forecast becomes a monthly settlement obligation.
Weak credit	Guarantee or collateral equal to 50 percent of total minimum charges for the full term, unless the customer meets specified credit and cash tests	A promise is moved up the hierarchy of money by backing it with credit support.
Stranded asset risk	Exit fee equal to 36 months of minimum charges after the allowed period	The customer pays to walk away from reserved capacity.

In Money View terms, AEP Ohio's tariff tries to move the survival constraint back onto the large customer. If the data center asks the grid to reserve capacity, the tariff makes that reservation a cash obligation, not just a growth forecast.

17.2 Virginia SCC and Dominion: a separate large-load class

Virginia is the center of the U.S. data center boom, so its regulatory decisions matter nationally. In Dominion Energy Virginia's 2025 biennial review, the Virginia State Corporation Commission approved a new GS-5 rate class for large-scale users, including customers with demand of 25 MW or more, effective January 1, 2027.[6]

The SCC's order and fact sheet describe several safeguards. Again, the important point is not whether one likes or dislikes data centers. The important point is that the regulator is translating a public promise into enforceable rate design.

Virginia is special because the issue is not one isolated building. It is a clustered hyperscale load pattern that affects long-term resource planning, transmission planning, and customer-class design. GS-5 matters because it pulls large loads out of ordinary rate classes and makes them visible as their own settlement category. But that does not end the inquiry. The 60 percent generation floor and 85 percent transmission and distribution floor still need to be tested against Dominion's IRP, resource-adequacy needs, and forecast-error scenarios. If generation and transmission buildout are justified by a systemwide data-center forecast, GS-5 may be a partial ring-fence rather than a complete settlement.

[6] Source: Virginia State Corporation Commission, 2025 Dominion Energy Virginia biennial review order announcement, listed in References.

Table 68: Virginia GS-5 provisions as Money View settlement devices

Risk	Regulatory device	Money View meaning
Cost shifting across customer classes	New GS-5 rate class for customers demanding 25 MW or more	Large-load costs are made visible in a separate ledger.
Reserved grid capacity not fully used	Minimum payment of 85 percent of contracted distribution and transmission demand, and 60 percent of generation demand for certain large-scale customers	The customer pays for capacity it causes the system to reserve, not only energy it actually consumes.
Long-lived infrastructure built for one class of load	Minimum 14-year take-and-pay obligation for new large-load customers contracting on or after January 1, 2027	The survival constraint is extended across the useful life of the grid investment.
Credit or nonperformance risk	Collateral obligation up to 60 percent of minimum charges for large-load customers without sufficient credit	The customer's promise is backed before losses can migrate to other customers.
Hidden allocation rules	Dominion must submit large-load interconnection processes and alternative cost allocation proposals	The settlement rule itself becomes a matter of public review.

This is the same ledger question in another form. The regulator is trying to keep infrastructure built for hyperscale loads from being recovered from other customers simply because the utility system sits in the middle.

17.3 What the two cases prove, and what they do not

These cases do not prove that AI data centers are harmless. They do not prove that every cost has been assigned correctly. They do prove something narrower and more useful: the public debate can move from ideology to settlement design.

Once minimum bills, collateral, exit fees, separate rate classes, and cost-allocation orders are on the table, the debate becomes concrete. Supporters can point to the provisions that protect households. Opponents can test whether those provisions are strong enough. Regulators can revise the settlement rule instead of arguing about whether the future is good or bad.

17.4 Applying the test to AEP Ohio: a public worked example

A full worked case would require project-specific load, cost, ramp, collateral, and credit data, some of which may be redacted or customer-specific. But AEP Ohio's public tariff terms are enough to show how the Ledger Adequacy Test works.

Table 69: Partial Ledger Adequacy Test for AEP Ohio's public tariff terms

Ledger item	Public tariff term	Valuation rule	Residual question
Queue and study risk	Load-study fee from \$10,000 to \$100,000	Counts as early cash, but only for study and queue discipline.	Does it cover planning time and opportunity cost if speculative projects crowd the queue?
Cancellation or delay before energization	Reimbursement of 100 percent of buildout costs if canceled or delayed more than 12 months before target energization	Strong for direct construction costs if buildout costs are defined and enforceable.	Does it include shared upstream planning, generator commitments, or RTO effects?
Underuse during ramp	Contract term tied to ramp period plus eight years, with minimum demand rules tied to contract capacity	Value depends on contracted capacity, billing determinants, and enforceable monthly collection.	What happens if actual usage reaches only 60 percent of forecast for multiple years?

Ledger item	Public tariff term	Valuation rule	Residual question
Credit recovery	Guarantee or collateral equal to 50 percent of total minimum charges unless credit and cash tests are met	Discount for credit quality, parent depth, callability, and priority against other creditors.	Does an exemption for strong credit leave households exposed if the payer is an affiliate or lease structure?
Early exit	Exit fee equal to 36 months of minimum charges after the allowed period	Counts as transition protection, not automatically full stranded-asset protection.	Is 36 months enough if the remaining net book value runs for decades?
Regional capacity and local fiscal costs	Not fully resolved by a retail distribution tariff alone	Apply jurisdiction discount: some exposure may sit in PJM, tax, water, or local infrastructure ledgers.	Which docket captures costs outside AEP Ohio's filed retail tariff?

Table 70: AEP Ohio Schedule DCT: public-data worksheet

Worksheet line	Public input from AEP Ohio	What the ledger can decide now
Project threshold	Data centers or expansions of 25,000 kW or greater must pay a load-study fee and follow the Schedule DCT process.	Small projects may follow a different path; large projects enter the special ledger.
Study fee	$10,000 for more than 25 MW and less than 50 MW; $50,000 for 50–100 MW; $100,000 for 100 MW and greater.	Queue discipline improves, but the fee is not large enough to cover major stranded infrastructure.
Ramp schedule	Contract capacity during ramp is at least 50% in year one, 65% in year two, 80% in year three, and 90% in year four.	Underuse risk is partly converted into a scheduled payment obligation before full load arrives.
Minimum demand arithmetic	For more than 75 MW, minimum demand is 57.5 MW plus 100% above 75 MW, capped at 85% of total contract capacity.	A 100 MW project has an 82.5 MW floor; 300 MW and 500 MW projects hit the 85% cap, or 255 MW and 425 MW.
Collateral	Unless credit and cash tests are met, guarantee or collateral equals 50% of total minimum charges for the full contract term.	Recovery value cannot be scored without payer identity, exemption status, form of collateral, and call conditions.
PJM forecast inclusion	AEP says Schedule DCT customer load will be included in its PJM forecast and transmission infrastructure will proceed through PJM planning.	The retail tariff does not by itself settle PJM forecast, capacity, RBP, or transmission-allocation exposure.
Exit and assignment	Capacity reduction can involve assignment to another Schedule DCT customer; exit fee is tied to minimum charges.	This may reduce stranded-asset risk, but public data do not show remaining net book value under year-5, year-10, or year-15 exit.

Table 71: What the public AEP Ohio record can and cannot decide

Ledger issue	Public status	Interpretation
Direct construction cancellation risk	Stronger than ordinary tariff treatment	Public terms appear to move direct buildout risk toward the customer if cancellation or delay occurs before energization.
Underuse during ramp	Conditional pass	Minimum demand rules help, but the pass depends on contracted MW, billing determinant, and actual ramp path.
Credit recovery	Conditional or unknown	Public terms identify collateral or guarantee requirements, but recovery value depends on payer identity, exemptions, callability, and priority.
Exit after several years	Unknown	A 36-month exit fee may be a transition payment rather than full remaining-net-book-value protection.
Regional PJM exposure	Not resolved by the retail tariff alone	AEP Ohio's tariff cannot by itself settle all PJM capacity, RBP, or transmission allocation questions.
Local fiscal and environmental ledger	Outside the tariff	Tax, water, roads, emissions, and school-district effects need separate public records.

This worked example does not say AEP Ohio's tariff fails. It says the tariff improves the ledger for direct construction, underuse, and credit risk, while leaving public questions about late exit, regional capacity spillovers, and non-electric public costs. That is the point of the test: it turns "good tariff" or "bad tariff" into a list of recoverable dollars, discounted promises, and residual claimants.

Table 72: AEP-style docket result template

Ledger	Provisional result	What must be filed next
Direct construction	Likely pass if public tariff terms are applied to defined buildout costs.	Itemized direct-facility cost and reimbursement trigger.
Underuse	Conditional pass.	Minimum-bill NPV under 60% realized load and delayed ramp.
Credit	Unknown pending payer and collateral review.	Legal payer, parent support, LOC or guarantee terms, callability, and priority.
Exit	Unknown pending net-book-value comparison.	Remaining NBV under year 5, 10, and 15 exit versus exit-fee NPV.
PJM/RTO	Unresolved by retail tariff alone.	Capacity obligation, RBP exposure, bilateral offset, and retail pass-through treatment.
Local ledger	Outside tariff.	Tax, water, roads, air, schools, and public-safety ledger.

17.5 Fully filled hypothetical worksheet: not AEP

The following example is not an estimate of AEP Ohio, Virginia, or any actual project. It shows how the scorecard should look when the public record is filled, discounted, and labeled. The point is not the exact dollar values; the point is that each line identifies exposure, instrument, credited value, gap, and residual claimant.

Table 73: Hypothetical 300 MW AI campus in a PJM state

Ledger line	Unit	Exposure	Instrument	Credited value	Gap	Residual claimant
Direct facilities	One-time capital exposure	$250M customer-specific facilities	Upfront contribution plus reimbursement agreement	$240M	$10M	Shareholder or ratepayers, depending on prudence review
Shared network	Assigned capital exposure	$150M upstream upgrade	Partial direct assignment and later-user credit	$80M	$70M	Regional customers unless later-user refund is enforceable

Ledger line	Unit	Exposure	Instrument	Credited value	Gap	Residual claimant
Underuse	Annual revenue shortfall	$99M/year fixed-cost target	85% minimum bill	$84M/year	$15M/year	Ratepayers or shareholders under rider true-up
Exit in year 5	Remaining net book value	$600M remaining NBV	36-month exit fee plus collateral draw	$120M	$480M	Ratepayers, replacement load, shareholders, or creditors
Capacity/RBP	Delivery-year or auction exposure	Unknown market exposure	No public bilateral or RBP charge	Unknown	Unknown	LSE and retail classes unless assigned
Water	Bond-term exposure	$80M water-authority bond	Water service agreement and minimum payment	$50M	$30M	Water users or local taxpayers
Local fiscal	NPV over abatement term	$120M abatement value; public-service costs unknown	PILOT and clawback	$60M	Unknown	County, schools, or residents

Calculation notes for the hypothetical worksheet

The figures are placeholders for method demonstration, not project estimates. The Unit column separates one-time capital exposure, annual revenue shortfall, remaining net book value, delivery-year exposure, bond-term exposure, and NPV over an abatement term. Direct facilities ($250M), shared network upgrades ($150M), water debt ($80M), and local fiscal exposure ($120M abatement value) are exposure inputs. Underuse ($99M/year) is an annual fixed-cost recovery target, not an NPV. Exit in year 5 ($600M) is remaining net book value exposure. Credited values are settlement resources after illustrative legal, timing, credit, priority, jurisdiction, and performance discounts. In a docket, each placeholder must be replaced by actual cost studies, depreciation schedules, discount rates, billing determinants, collateral documents, payer identity, and replacement-load assumptions.

Hypothetical result

Direct electric ledger: conditional pass. Underuse ledger: conditional pass. Exit ledger: fail under year-5 stress unless additional collateral or assignment is filed. Regional capacity ledger: unknown. Local water and fiscal ledgers: conditional or unknown. Overall result: no finding of subsidy, but public cost recovery is not justified until the unknown and negative ledgers are closed.

17.6 Docket-ready scorecard for a full worked case

The next step is to run the same test through a complete public docket. If customer-specific data are redacted, the docket can still use ranges, proxy evidence, and confidential review by staff or public counsel. The point is to identify exactly where the public record is complete, where it is sealed but reviewable, and where it is missing.

Table 74: Full Ledger Adequacy Test template for one public docket

Ledger line	What the docket must supply	Public or proxy source	Residual claimant if missing
Direct interconnection	Customer-specific facilities, cost estimate or range, upfront payment, reimbursement rule, and service date	Tariff, service agreement, interconnection study, staff testimony	Other utility customers or utility shareholders
Shared network upgrades	Which upgrades are direct assigned, which are shared, and whether later users provide refunds or credits	Rate case, transmission study, RTO filing, cost-allocation order	Regional customers, future customers, or later interconnecting users

Ledger line	What the docket must supply	Public or proxy source	Residual claimant if missing
Rate-base exposure	Assets entering rate base, assets paid upfront, carrying charge, depreciation life, and rider or base-rate recovery path	Rate case, IRP, certificate proceeding, utility capex guidance	Ratepayers if not ring-fenced; shareholders if disallowed
Exit scenario	Remaining net book value under exit years 5, 10, and 15 compared with exit-fee NPV	Tariff, depreciation schedule, cost-of-service study, public counsel analysis	Ratepayers, replacement customers, or shareholders
Credit scenario	Parent guarantee, letter of credit, deposit, escrow, collateral amount, callability, priority, and legal payer	Credit-support agreement under protective order, bond disclosure, utility testimony	Utility, ratepayers, secured creditors, or bankruptcy estate
PJM or RTO exposure	Capacity obligation, LSE pass-through, retail rider treatment, and any customer-specific offset or qualifying capacity contribution	RTO auction data, retail tariff, FERC filing, LSE testimony	Regional customers or retail classes without customer-specific assignment
Local ledger	Tax abatement, water, roads, public safety, school-district effects, emissions, noise, land use, and clawbacks	County agreement, fiscal note, water authority bond disclosure, zoning record	Taxpayers, school districts, water users, or nearby residents
Residual claimant	Final payer if each instrument fails under delay, underuse, exit, default, technical nonperformance, or forecast error	Ledger adequacy certificate by staff, public counsel, or consumer advocate	The party named in the scorecard, not the party named in the press release

17.7 Stress-testing the safeguards

The real Money View test is not whether a tariff contains protective words. It is whether those words hold when the payment chain is stressed.

Table 75: Stress tests for AEP Ohio and Virginia-style large-load tariffs

Stress scenario	What the current tools may cover	Open question
Project cancellation before energization	Study fees, buildout-cost reimbursement, collateral, and exit rules can push direct construction risk back to the customer.	Does the reimbursement cover shared upstream planning costs, generator commitments, and regional market effects?
Load reaches only 60 percent of forecast	Minimum demand charges such as 85 percent T&D floors or 60 percent generation floors reduce the underuse gap.	Why is the chosen percentage sufficient for the asset actually built, rather than merely politically acceptable?
Exit after several years	Exit fees and reassignment provisions can reduce stranded-asset exposure.	Is a 36-month exit fee enough for assets with 20- to 40-year lives, or does it only cover a transition period?
Credit deterioration or bankruptcy	Parent guarantees, letters of credit, deposits, and collateral requirements improve enforceability.	Is the guarantee senior to other claims and enforceable against a deep parent, or limited to a thin project company?
Project is split across affiliates or phases	Aggregation rules and threshold definitions can prevent obvious evasion.	Does the 25 MW or similar threshold capture staged campuses, behind-the-meter structures, and affiliated entities?
Regional capacity prices rise	A state tariff can assign local grid costs and minimum bills.	Can it capture RTO-wide capacity spillovers, or does that require regional market-rule reform?
Local tax revenue underperforms	Clawbacks and public-benefit agreements can recover some public concessions.	Are school districts, water systems, road agencies, and emergency services protected, or only the electric utility?

This stress test turns the tariff from a policy label into a living balance sheet. A strong provision is one that still identifies a payer after cancellation, delay, underuse, credit deterioration, market spillover, and political disappointment.

18 Evidence Ladder: How to Tell Who Is Really Paying

Public debates often treat announcements as proof. A Money View approach needs an evidence ladder. Strong evidence changes the ledger. Medium evidence points to a likely direction. Weak evidence shows that a story exists, but not that a payment obligation exists.

There is one complication: the best contracts are often partly redacted. Interconnection agreements, special contracts, and customer-specific service terms may be protected as commercially sensitive information. That does not mean the public is helpless. It means the evidence ladder must include procedural tools and proxy evidence.

Because information is asymmetric, the evidence burden should be asymmetric too. If a project seeks rate recovery, tax incentives, public infrastructure, or expedited approvals, the applicant and utility should disclose enough information for the public to test cost causation. Residents should not be asked to prove a subsidy from behind a wall of redactions.

Table 76: Evidence ladder for data center cost shifting

Evidence strength	What to look for in the United States	How the public can get closer to it
Strong evidence	Executed large-load tariff, interconnection agreement, rate-case order, capacity-market settlement data, minimum-bill clause, collateral requirement, exit-fee provision, or unredacted testimony available to commission staff, public counsel, or a consumer advocate under protective order.	Search the commission docket, rate case, tariff docket, IRP docket, and public counsel filings. Watch cross-examination, settlement testimony, staff reports, and commission orders for the numbers that survive redaction.

Evidence strength	What to look for in the United States	How the public can get closer to it
Proxy evidence	Utility capital-expenditure guidance, earnings-call transcripts, investor presentations, credit-rating reports, bond disclosures, quarterly large-load queue reports, and data-center load assumptions in IRPs.	Use SEC filings, investor-relations materials, earnings-call transcripts, municipal bond disclosures, and required utility queue reports. These sources often reveal the scale, timing, and risk allocation even when the customer contract is redacted.
Medium evidence	Utility load forecast, integrated resource plan, regulatory hearing record, fiscal impact analysis, developer commitment to grid upgrades, proposed rider, proposed certificate of need, or proposed generation addition.	Track IRP proceedings, certificate proceedings, rider filings, and local-government fiscal analysis. Ask whether the proposal includes multiple load-growth scenarios and customer-specific cost assignment.
Weak evidence	Company press release, economic-development speech, vague ”we will pay our own way” statement, broad clean-energy claim without contract detail.	Treat it as a lead, not proof. Use it to identify the utility, docket, project site, load size, expected service date, and promised public benefits.

The test is simple. If a project says it will not burden households, ask whether that promise appears in a tariff, contract, budget, regulatory order, public counsel filing, or credible proxy record. If it does not, the claim is still a story.

Table 77: Source-pinning register for formal release

Claim to pin	Required pin	Current source type	Status before filing
IEA global data-center demand, 2024 and 2030	IEA *Energy and AI* executive summary, "Data centres account..." and "Electricity demand..." sections for 415 TWh and 945 TWh.	Official intergovernmental report.	Citation-locked to web section; add PDF page if filing requires page pin.
LBNL U.S. data-center demand, 2023 and 2028 scenarios	LBNL *2024 United States Data Center Energy Usage Report*, executive summary page 6 for 176 TWh and 325–580 TWh; report page 51 / Figure 5.5 for the same range.	National laboratory report.	Citation-locked to report pages; add table/figure extract if filing.
Monitoring Analytics $16.6B and $23.1B counterfactuals	2026/2027 Part A, pages 2–3, for $7.271B and $16.603B; 2027/2028 Part A, pages 20 and 23 / Table 11 for updated auction components.	Market monitor analysis.	Citation-locked to report pages/tables; verify arithmetic before filing.
AEP Ohio Schedule DCT terms	AEP Ohio official DCT page, "Important Tariff Information" for study fees, load ramp, collateral, minimum demand, and 36-month exit fee.	Utility tariff/public tariff summary.	Citation-locked to official page sections; add tariff sheet/-docket if filing.
Virginia GS-5 terms	SCC Nov. 25, 2025 release for 25 MW, 85 percent T&D, 60 percent generation, and Jan. 1, 2027 effective date; SCC facts page for 14-year take-and-pay, collateral, and interconnection-process review.	State commission release and official SCC facts page.	Citation-locked to official SCC pages; add final order paragraph/page before formal legal filing.
PJM RBP and connect-and-manage status	PJM Feb. 18, 2026 working paper page 4 for inability to assign costs directly to specific loads; PJM Apr. 16, 2026 design presentation pages 8–10 and 20–21 for targets and EDC allocation; PJM Inside Lines for stakeholder/status context.	RTO materials and journalism for evolving status.	Citation-locked to RTO materials as of May 22, 2026, 19:14 EDT; update before filing or citation.

19 Procedural Remedies When the Ledger Is Hidden

A public ledger test only works if the institutions with jurisdiction can see the ledger. Many data-center contracts, interconnection studies, credit arrangements, and special service terms contain commercially sensitive information. That is a reason for controlled disclosure, not a reason for public blindness.

Table 78: Procedural remedies for hidden ledgers

Problem	Remedy	Public purpose
Project seeks rate recovery, tax incentives, public infrastructure, or expedited permitting	Create a disclosure presumption: the applicant, utility, or public agency must produce enough information to run the ledger test.	The burden of proof stays with the party asking for public support or cost recovery.
Commercially sensitive contracts are redacted	Permit confidential filing under protective order for commission staff, public counsel, consumer advocates, or other statutory representatives.	Sensitive price and identity data can be protected while the public's representative tests the numbers.
Applicant refuses cost-causation data	Recommended default for public-cost recovery: no ordinary rate recovery, public subsidy, or public-risk shifting absent an adequate evidentiary showing, unless the decision-maker makes a supported finding that the cost is just, reasonable, and properly allocated.	A hidden ledger should not become a public bill.
Special contracts remain confidential	Require a public non-sensitive summary: MW, term, service date, minimum bill, exit fee, collateral type, payer identity category, and cost categories covered.	The public can see the settlement architecture even if exact commercial terms remain sealed.
The record is too technical for residents to audit	Require public counsel, a consumer advocate, or commission staff to file a ledger adequacy certificate.	A responsible institution states whether enforceable settlement resources cover downside exposure.

The procedural rule is simple: recommended default for public-cost recovery is no ordinary rate recovery, public subsidy, or public-risk shifting absent a public-facing ledger and an adequate evidentiary showing. If the project wants ordinary customers, taxpayers, or public agencies to stand behind the buildout, the applicant should show how those public balance sheets are protected.

Table 79: Interim approval options when the ledger is incomplete

Interim option	Deadline and evidence condition	Reopener, true-up, and carrying-cost rule
Conditional approval with escrow	Approval expires or reopens on a set filing date unless escrow, LOC, or bond covers the disputed exposure.	Automatic true-up when the ledger is filed; interim carrying cost assigned to escrow, customer, or shareholders as ordered.
Construction allowed, no general rate recovery yet	Utility may build customer-specific facilities, but general rate recovery waits for a Ledger Adequacy Certificate by a date certain.	Reopener if the certificate is missing or negative; carrying cost deferred, customer-specific, or shareholder-shared until supported recovery.
Provisional forecast treatment	Forecast inclusion requires executed agreement, milestone deposit, cancellation payment, and next-update review date.	If milestone fails, remove or discount load and true up any committed procurement cost to customer collateral or approved allocation.
Time-limited confidentiality	Seal lasts only through a defined review window; staff, public counsel, or consumer advocate receives protected access.	If public summary or representative review is not filed by deadline, no final public-cost-recovery finding.
Emergency procurement with true-up	Reliability procurement may proceed for a defined emergency term with later cost-allocation evidence.	Automatic reopener and true-up to cost-causing load, LSE/EDC, customer class, or supported public-welfare finding.
Shareholder deferral	Utility may defer carrying costs until verified customer commitment or used-and-useful finding.	If evidence never arrives, commission decides disallowance, customer draw, shareholder sharing, or later-user reassignment.

Five procedural rules

1. No public-cost recovery or public-risk shifting after a public-risk trigger without ledger disclosure.
2. No confidentiality without public representative review.
3. No forecast inclusion without a meaningful and verifiable customer commitment.
4. No flexibility credit without telemetry, dispatch rights, and penalties.
5. No cross-ledger offset without an enforceable transfer mechanism.

19.1 Legal hooks for the ledger

The Cross-Jurisdiction Ledger Statement is useful only if a public institution can require it. Different institutions have different hooks.

Table 80: Legal hooks for requiring ledger disclosure

Institution	Possible hook	What it can require
State PUC	Rate recovery, certificate approval, large-load tariff, special contract, rider, or prudence review	Customer-specific cost assignment, minimum bills, collateral, service term, load forecast support, and ratepayer ring-fencing.
County or city	Tax abatement, PILOT, zoning approval, infrastructure support, or expedited permitting	Fiscal impact statement, clawbacks, water and road commitments, public-safety costs, and local benefit reporting.
Water authority	Water service agreement, infrastructure bond issuance, or capacity reservation	Project-specific rate, minimum water payment, drought-year limits, debt-service protection, and exit payment.
Environmental regulator	Air permit, backup generation permit, water withdrawal permit, or mitigation condition	Emissions limits, operating-hour limits, monitoring, neighborhood exposure, and mitigation funding.
RTO or FERC	Load forecast verification, interconnection rules, backstop allocation, transmission planning, or capacity-market participation	Verifiable load commitment, ramp schedule, duplicative-request checks, bilateral capacity documentation, and curtailment rules.
Consumer advocate or public counsel	Protective-order participation in PUC, rate, IRP, or special-contract proceedings	Review of redacted contracts, credit support, cost studies, and ledger adequacy certificates on behalf of the public.

Table 81: Institutional remedy boundaries: orders and conditions

Institution	Can order	Can condition
State PUC	Retail rate recovery, tariff approval, riders, prudence findings, and customer class design where authorized.	Ring-fencing, collateral, minimum bills, service terms, true-ups, forecast support, and public summaries.
County or city	Tax abatements, PILOTs, zoning, road deals, local infrastructure support, and local clawbacks.	Public-benefit agreements, school hold-harmless terms, road payments, infrastructure milestones, and reporting duties.
RTO or FERC	Forecast rules, interconnection terms, transmission planning, capacity-market rules, and tariff-approved allocation mechanisms.	Bilateral capacity paths, curtailment accreditation, deliverability requirements, and load-commitment verification.
Water authority	Water service, project-specific rates, capacity reservation, debt protection, and exit payments.	Drought limits, minimum payments, infrastructure milestones, and stranded-debt protection.
Environmental or land-use agency	Permit limits, monitoring, mitigation, zoning conditions, and enforcement procedures within its statute.	Backup-generator operating limits, noise buffers, emissions controls, and mitigation funds.

Table 82: Institutional remedy boundaries: disclosure and referral

Institution	Can request or disclose	Must refer elsewhere
State PUC	Local fiscal summary, water-service status, affiliate payer identity, and RTO pass-through treatment relevant to rate recovery.	County tax abatements, zoning, air permits, and local water debt outside the PUC's authority.
County or city	Utility/RTO status, power-service readiness, water debt exposure, and public-cost summary.	Retail rate allocation, PJM capacity assignment, utility prudence, and FERC/RTO tariff design.
RTO or FERC	Load commitment data, ramp schedule, duplicative queue requests, and large-load market participation evidence.	County fiscal benefits, local water and land-use burdens, and final retail class billing unless passed through state proceedings.
Water authority	Electric-service readiness, cooling assumptions, public debt term, and project cancellation provisions.	PUC retail rates, RTO capacity allocation, air permits, and tax abatements.
Environmental or land-use agency	Power, water, traffic, and emergency-service assumptions relevant to site impacts.	Utility cost recovery, parent guarantees, and RTO market charges.

Table 83: Three versions of the Cross-Jurisdiction Ledger Statement

Version	What it should cover	Who can usually ask	What it cannot solve alone
PUC version	Rate recovery, tariff, service agreement, cost-to-serve study, credit support, minimum bill, exit fee, load forecast, and ratepayer ring-fencing.	State PUC, staff, public counsel, utility, intervenors where authorized.	County tax deals, water debt, air permits, and RTO-wide allocation unless linked to the PUC docket.
Local government version	Tax abatement, PILOT, water, roads, schools, public safety, land use, noise, local fiscal NPV, and clawbacks.	County, city, zoning board, water authority, school district, local public finance body.	PJM/RTO capacity costs, retail class billing, utility rate base, and parent guarantees outside local agreements.
RTO/FERC version	Forecast verification, capacity allocation, transmission cost allocation, curtailment accreditation, bilateral capacity contracts, and large-load market participation.	RTO, FERC, market monitor, state PUC through pass-through review, LSE/EDC filings.	County fiscal benefits, local water debt, site-level environmental burdens, and retail distributional effects unless passed through state proceedings.

Table 84: Prototype authority memos: legal hook, order power, condition power, referral

Prototype	Legal hook	Can order	Can condition	Must refer elsewhere
Ohio / AEP-style PUC memo	Large-load tariff approval, retail rate recovery, rider treatment, prudence review, and state pass-through proceedings where authorized.	Filed tariff terms, minimum-bill structure, collateral disclosure, cost-to-serve evidence, customer-specific recovery treatment, and public summary of redacted terms.	Rate-base/rider recovery on milestone deposits, executable service agreements, exit fees, ring-fencing, shareholder sharing, or Ledger Adequacy Certificate.	County tax deals, local water debt, air permits, and PJM/FERC tariff design.
Virginia / Dominion-style IRP memo	IRP review, biennial/rate case, rider review, large-load class design, certificate or prudence proceedings where authorized.	Forecast scenarios, GS-5 class treatment, generation and T&D floor analysis, rate-base/rider exposure, 14-year obligation stress test, and public summary.	Resource-plan or recovery approval on signed load commitments, generation-floor adequacy, used-and-useful review, true-ups, or customer-specific risk sharing.	County abatements, site permits, PJM/FERC wholesale rules, and local water/fiscal remedies outside SCC authority.
PJM / FERC memo	RTO tariff, FERC filing, forecast verification, capacity-market rules, interconnection process, RBP design, and curtailment accreditation.	Forecast-verification data, large-load commitment evidence, market participation rules, capacity allocation mechanics, and telemetry/accreditation standards.	RBP participation, bilateral capacity path, connect-and-manage rights, qualifying capacity credit, and curtailment value on verifiable commitments.	Retail class billing, county fiscal benefits, local water debt, zoning, and state affordability mitigation.

Table 85: Sample order conditions

Condition	Sample language
Ledger certificate	Recommended condition: customer-specific facility costs should not enter general rates unless the applicant files a Ledger Adequacy Certificate identifying payer, instrument, collateral, timing, and residual claimant, where authorized by law.
Forecast support	Recommended condition: forecast load should not support rate-base recovery unless tied to an executed service agreement, deposit, milestone schedule, or equivalent enforceable commitment.
Flexibility credit	Recommended condition: any claimed flexibility credit should be contingent on telemetry, dispatch rights, test events, and nonperformance penalties sufficient to cover avoided capacity value and reliability externality.
Confidentiality summary	Recommended condition: confidential commercial terms may be sealed, but payer identity category, MW, term, cost categories covered, collateral type, and residual claimant should be summarized publicly.
Cross-ledger offset	Recommended condition: tax, water, workforce, or local-development benefits should offset electric or regional-market exposure only if an enforceable transfer mechanism moves cash, credit, capacity, or legal responsibility to the affected ledger.

Table 86: Sample order packages by institution

Institution	Sample package
PUC	Ledger Adequacy Certificate; cost-to-serve study; minimum-bill and exit-fee NPV; collateral scope; forecast milestone schedule; public summary of redacted terms.
County or city	Net fiscal statement; abatement and PILOT terms; school hold-harmless analysis; road and public-safety costs; clawback and reporting requirements.
Water authority	Project-specific water agreement; average and peak withdrawals; debt-service protection; drought-year limits; minimum payment and exit payment.
RTO or FERC	Forecast verification; load commitment evidence; capacity or transmission allocation status; bilateral capacity path; curtailment accreditation and telemetry requirements.
Public counsel or consumer advocate	Protective-order access to service agreements, credit support, cost studies, sponsor-provided data, and draft ledger certificate; public non-sensitive summary.

20 Local Fiscal and Environmental Ledger Appendix

The local ledger should be calculated, not narrated. A county press release may report capital investment and jobs, while the school district, water authority, road agency, and nearby residents hold separate claims. The minimum public record should include the following entries.

This is not a side issue. Associated Press reporting on the 2026 ratepayer pledge noted that community backlash has focused not only on electricity prices, but also on pollution and water consumption. A public ledger that stops at the electric tariff will miss much of what local residents are actually contesting.

Table 87: Required local ledger data

Ledger item	Required data	Public test
Taxes and incentives	Abatement term, annual tax before and after abatement, NPV, school-district loss, PILOTs, and clawback triggers	Does the net tax benefit remain positive after public concessions?
Water and wastewater	Average and peak daily withdrawal, cooling technology, drought-year limits, public capex, debt term, debt service, project-specific rate, cancellation rule, and stranded-asset treatment	Is water infrastructure paid by the project, the water authority, or existing users if demand fails to appear?
Cooling scenario	Water-cooled, air-cooled, hybrid, or recycled-water design, with power-use tradeoff under heat and drought scenarios	Does a water-saving design raise electric peak demand, or does an efficient electric design raise water stress?
Backup generation and air permits	Diesel or gas MW, operating-hour limits, emissions controls, expected run hours, reporting rules, mitigation cost, and nearby population exposure	Are reliability benefits being bought with local air-quality costs?
Roads, fire, and public safety	Capital upgrades, staffing, response-time impact, fire suppression needs, and annual operating cost	Do emergency and transportation costs appear in the fiscal deal?
Land use, noise, and mitigation	Zoning conditions, buffers, noise limits, mitigation fund, and enforcement process	Are nearby communities compensated for site-specific burdens?
Transmission siting	Easements, route alternatives, compensation, affected communities, and benefit allocation	Are land burdens concentrated while regional benefits are dispersed?

The same appendix should include a calculation template, not only a list of issues. The local claim is credible only when the public can see the baseline, the project case, the discount rate, and the residual payer.

Table 88: Local ledger NPV template

Line item	NPV inputs	Residual claimant question
Property tax	Baseline tax without abatement, tax after abatement, term, discount rate, assessment assumptions, and school-district split	Which public entity loses revenue if the assessed value, tax rate, or project life disappoints?
PILOTs and fees	Annual amount, escalation, conditions, payment priority, default rule, and clawback trigger	Are payments enforceable if the project is delayed, sold, or restructured?
Water and wastewater	Average and peak withdrawal, capex, debt term, debt service, project-specific rate, ratepayer allocation, drought-year constraints, and operating costs	Do existing water customers carry debt for facilities built around the data center?
Water stranded-asset case	Remaining bond principal if the project cancels or underuses water service, plus any customer minimum payment or exit fee	If the data center leaves, who pays the water authority's remaining debt service?
Roads, fire, and public safety	Upfront capital cost, annual staffing, equipment replacement, emergency-response standard, and funding source	Does the county budget absorb recurring service costs after the ribbon-cutting?
Backup generation	Permit limit, expected operating hours, emissions controls, mitigation cost, monitoring, enforcement cost, and nearby population exposure	Who pays for local air-quality risk created by reliability planning?
Noise and land use	Mitigation fund, monitoring cost, buffer rules, enforcement process, and affected households	Are nearby residents compensated or merely asked to absorb site-specific burdens?
Transmission siting	Easement compensation, affected households, route alternatives, mitigation, and benefit allocation	Are land burdens local while grid benefits and revenues are regional?

The local ledger does not replace the electric ledger. It prevents a project from looking clean in one ledger while shifting costs to another.

Table 89: Three local burden categories

Burden type	Examples	Required treatment
Cash-settled local costs	Water debt, wastewater capex, roads, fire staffing, school revenue, public-safety equipment.	Quantify in dollars or NPV; identify payer, instrument, and residual claimant.
Permit-governed burdens	Air permits, backup generation, water withdrawal limits, operating-hour limits, stormwater, land-use conditions.	Disclose limits, monitoring, enforcement, mitigation, and agency jurisdiction.
Non-monetized community burdens	Noise, land-use concentration, nearby resident exposure, procedural exclusion, visual and siting burdens.	Do not force every burden into dollars. Require disclosure, mitigation, monitoring, enforcement, affected-community benefit agreement, or explicit public-welfare finding.

Table 90: Three layers of local governance

Layer	What it asks	Why it matters
Legal compliance layer	Has the project satisfied zoning, air, water withdrawal, backup-generation, road, stormwater, and other minimum permit requirements?	Compliance establishes legality, not necessarily who bears public cost or concentrated burden.
Settlement layer	Are quantifiable local costs assigned to a legal payer through PILOT, clawback, water rate, minimum payment, road agreement, school hold-harmless, or public-safety funding?	Cash-settled public costs need a payer, instrument, time horizon, and residual claimant.
Community mitigation layer	Are non-monetized or concentrated burdens disclosed, monitored, mitigated, compensated, or addressed through an affected-community benefit agreement?	Some burdens should not be forced into NPV, but they still require record evidence and enforceable conditions.

Permit compliance is not community settlement

An air permit, water-withdrawal permit, backup-generation permit, or zoning approval can show legal compliance with a minimum regulatory standard. It does not automatically show that nearby residents, renters, schools, water users, or affected communities have been compensated, protected, or represented. Permit compliance belongs in the local ledger; it is not the same as local settlement.

Local nexus rule

Do not let the local ledger become a general forum for every political objection. A local burden should enter the ledger only after the record identifies: (1) the causal nexus to the project, cluster, queue, or forecast class; (2) the affected group; (3) the mitigation, payment, monitoring, or disclosure pathway; and (4) the institution with authority to act. If one of those elements is missing, the issue may still matter politically, but it should not be counted as a settled ledger line.

20.1 Distributional ledger

The public ledger is not only a total-dollar question. It is also a distributional question: who receives the benefit, who pays the bill, who cannot exit, and who cannot negotiate the contract.

Table 91: Distributional ledger

Distributional question	Evidence to request	Why it matters
Low-income bill burden	Bill-impact model by income class, arrearage data, energy burden, and low-income protections.	A small average increase can be material for households near affordability limits.
Small business exposure	Class allocation, demand-charge treatment, rider exposure, and commercial customer impacts.	Small businesses may not receive large-load benefits but may share fixed-cost recovery.
School-district fiscal effect	Tax abatement split, school funding formula, PILOT allocation, and clawback rules.	A positive county ledger can coexist with a negative school ledger.
Water users vs. electric ratepayers	Water debt service, project-specific water rate, and overlap with electric customer classes.	The group paying water debt may not be the group receiving electric benefits.
Nearby residents	Noise, backup emissions, transmission route, land-use buffers, and mitigation funds.	Site burdens may concentrate in communities that do not share project gains.
Municipal and cooperative customers	Wholesale pass-through, contract exposure, and local board approval records.	Public-power or co-op customers can face different risk paths from IOU customers.
Renters	Utility bill exposure, housing pass-through, and lack of property-tax benefit.	Renters may absorb bill increases without receiving property-tax gains.
Procedural access	Notice, language access, intervention cost, public counsel participation, and protective-order access.	A ledger cannot depolarize a process that affected communities cannot enter.

Distributional Adequacy Rule

A project may pass aggregate settlement review and still require mitigation if material costs fall on customers or communities that do not receive corresponding benefits, cannot exit, and were not represented in the negotiation. Distributional review does not mean every local burden blocks a project. It means the record must name who pays, who benefits, who cannot avoid the burden, and which mitigation or compensation instrument follows.

Table 92: Distributional materiality triggers

Burden	Trigger for mitigation review	Typical remedy path
Low-income bill burden	Bill impact materially increases energy burden for protected or low-income classes, or exceeds an affordability-program threshold.	Bill credit, arrearage protection, affordability fund, or shareholder/customer-specific contribution.
Small business exposure	Class allocation shifts fixed costs or demand-charge exposure materially to small commercial customers without offsetting benefit.	Class-specific true-up, rider design change, or exclusion from ordinary pass-through.
School fiscal effect	Abatement or PILOT allocation produces school revenue loss above budget, per-pupil, or hold-harmless threshold.	School hold-harmless, PILOT allocation, clawback, or explicit welfare finding.
Water users	Water debt service or stranded capacity risk falls on existing users and project minimum payment does not cover bond-term exposure.	Project-specific water rate, debt-service reserve, minimum payment, or exit payment.
Nearby residents	Noise, backup emissions, transmission siting, traffic, or land-use burden reaches permit or mitigation trigger.	Monitoring, operating limits, mitigation fund, buffer, or community benefit agreement.
Procedural access	Notice, language access, intervention funding, public counsel review, or protective-order access is missing for affected groups.	Delayed final approval, public summary, public counsel funding, or access remedy.

Threshold-source rule

"Material" should be tied to a jurisdiction-specific source where possible: affordability-program or energy-burden standards for low-income customers; class-allocation or demand-charge thresholds for small business; per-pupil, annual-budget, or hold-harmless thresholds for schools; bond-term debt-service coverage for water users; permit mitigation triggers, noise standards, generator-hour limits, emissions limits, traffic thresholds, or siting conditions for nearby residents; and notice, language-access, public-counsel, or protective-order access rules for procedural adequacy.

Table 93: Distributional remedies

Material burden	Possible mitigation
Low-income electric bill exposure	Bill credits, arrearage protection, affordability rider, or shareholder/customer-specific contribution to a utility affordability fund.
School-district fiscal loss	Hold-harmless payment, PILOT allocation to schools, clawback trigger, or state funding adjustment disclosure.
Water-user stranded debt	Project-specific water rate, minimum water payment, debt-service reserve, exit payment, or later-user reimbursement mechanism.
Noise, emissions, or backup-generator burden near the site	Mitigation fund, operating-hour limits, monitoring, enforceable response protocol, buffer, or affected-community benefit agreement.
Renter exposure without property-tax benefit	Utility affordability fund, landlord pass-through disclosure, local housing mitigation, or targeted bill support.
Procedural exclusion	Public counsel funding, language access, notice improvements, protective-order review, and plain-language ledger summary.

21 What Would Change Our Mind?

The ledger test must be falsifiable. If the evidence shows that a data center covers the costs it causes and contributes surplus revenue to the system, opponents should acknowledge that the electric ledger has passed.

This is not hypothetical. E3's Amazon-commissioned study compared utility revenues from evaluated Amazon data centers with estimated cost to serve in four utility territories. It concluded that those facilities generated sufficient or surplus net revenue in the studied cases, with an assumed 100 MW data center producing about $3.4 million in surplus utility revenue in 2025 and up to $6.1 million in 2030 under the study's assumptions. That kind of result is a serious supporter countercase.[7]

It is also not a blank check. The report itself says it was funded by Amazon, reviewed by Amazon before publication, and based on data provided by Amazon plus public sources where available; it also says E3 is an independent consulting firm and that its conclusions are its own. That places the report in the evidence ladder as a serious but interested countercase. A facility-level surplus study may depend on the specific tariff, local cost structure, load profile, customer credit, and scope of costs included. It may show that the data center covers its direct utility cost while leaving regional capacity, water, tax, land-use, or forecast-error costs outside the frame. The right response is not to dismiss the study or accept it uncritically. The right response is to ask whether the full ledger passes.

What would make a project pass the ledger test?

- The executed tariff or contract assigns direct and shared grid costs to the large-load customer.
- A cost-of-service or revenue-cost study shows no shortfall under low-load and delayed-ramp scenarios.
- Collateral, parent guarantees, minimum bills, and exit fees cover remaining net book value under default or early exit.
- Capacity-market impacts are separately allocated, offset by customer-provided qualifying capacity, or addressed through RTO rules.
- The local fiscal analysis remains positive after abatements, school-district effects, water and road costs, emergency services, and environmental mitigation.

[7]Source: Energy and Environmental Economics, *Tailored for Scale: Designing Electric Rates and Tariffs for Large Loads*, listed in References.

- Legal payers across electric, fiscal, water, and land-use ledgers are either the same creditworthy entity or linked by enforceable guarantees, cross-default provisions, collateral, or parent support sufficient to prevent strategic separation of liabilities.

If those conditions are met, the project should not be described as a public subsidy simply because it is large. A large load can be risky and still be well priced. It can stress the grid and still pay for the stress it creates. The public standard should be evidence, not suspicion.

What would make supporters lose fairly?

Supporters should concede public risk remains unresolved if the legal payer is unclear; the minimum bill does not cover fixed-cost exposure; exit fee plus collateral falls short of remaining net book value; PJM or RTO exposure has no enforceable assignment mechanism; local fiscal claims omit water, schools, roads, emergency services, or mitigation; parent support covers only direct bills rather than shared obligations; or confidential terms cannot be reviewed by staff, public counsel, consumer advocate, or another public representative under protective order.

22 Depolarization Research Appendix

This toolkit is a depolarization hypothesis as well as an audit method. The hypothesis is that ledger framing can move a public meeting from identity conflict to evidence conflict: not "AI good" versus "AI bad," but "which contract pays which bill under which stress case?"

Table 94: How to test whether ledger framing reduces polarization

Research step	Method	Output
Classify public objections	Code public comments, news stories, and hearing testimony into electric bill, water, noise, pollution, tax abatement, secrecy, land use, trust, and jobs categories.	Shows whether opposition is ideological or tied to specific ledgers.
Compare framings	Survey experiment: one group reads "AI jobs vs environmental risk"; another reads an asset-bill-backstop ledger statement.	Measures support, opposition, trust, perceived fairness, and disclosure demand.
Track evidence response	Present both supporters and opponents with pass/fail ledger evidence.	Tests whether both sides update when contracts and residual claimants are visible.
Audit process trust	Measure whether protective-order review by public counsel increases perceived legitimacy.	Tests whether controlled disclosure can reduce suspicion without exposing sensitive terms.

Table 95: Minimal empirical pilot

Pilot element	Minimum design
Comment sample	Code 30–50 county-meeting comments, local news quotations, PUC comments, or written submissions from one or two live data-center controversies.
Coding scheme	Electric bill, water, noise, pollution, tax abatement, secrecy, land use, trust, jobs, national competition, and corporate accountability.
Ledger-specific measure	Identify whether each comment raises a concrete ledger item or a generalized pro/anti-AI identity claim.
Survey vignette	Compare a jobs-versus-risk frame with an asset-bill-backstop ledger statement.
Outcome measures	Support, opposition, trust, perceived fairness, willingness to accept evidence, and demand for contract disclosure.
Use of result	Treat the depolarization claim as preliminary until the ledger frame demonstrably improves evidence responsiveness across supporters and opponents.

Until such evidence exists, the depolarization claim should be stated modestly: ledger language may make disagreement more testable, but it has not yet been shown to dissolve political conflict by itself.

23 Where This Framework Could Be Wrong

A serious ledger test should be able to criticize itself. This framework could overstate cost-shift risk in several ways.

Table 96: How the framework could overstate risk

Possible error	How to correct for it
It may undercount system benefits from high-load-factor customers.	Require cost-to-serve studies and revenue-cost comparisons, then credit enforceable surplus margin after stress testing.
It may over-assign shared infrastructure costs to the first mover.	Use later-user credits, refund mechanisms, beneficiary analysis, and shared network allocation rules.
It may treat forecast error as customer-caused when broader electrification, retirements, or reliability upgrades also drive need.	Run counterfactual load scenarios with and without the data center, and assign only incremental exposure.
It may undervalue future beneficiaries of transmission or generation assets.	Require future-use scenarios, reassignment rules, and declining minimum bills when assets become broadly useful.
It may over-haircut parent guarantees where legal instruments are unusually strong.	Replace default haircuts with instrument-specific evidence on governing law, callability, priority, and parent balance sheet.
It may undervalue verified flexibility from workloads that are genuinely dispatchable.	Require telemetry, test events, and penalties, then credit accredited flexibility value rather than treating all data centers as rigid loads.

The remedy is not to abandon the ledger test. It is to require both costs and benefits to pass the same evidence and stress standards.

24 Five Takeaways

1. The grid question is not just energy. It is capacity, location, and settlement. AI data centers raise questions about peak demand, transmission bottlenecks, backup resources, capacity markets, IRPs, rate-base additions, and local reliability. Annual electricity use is only the first number. The sharper question is who must settle the bill when the system reserves capacity for a load that may or may not arrive as forecast.

2. "Big Tech will pay" must be translated into contracts and numbers. The public should look for minimum bills, upfront contributions, collateral, exit fees, clear responsibility for interconnection costs, interruptible-service rules, and protections against stranded assets. Then it should ask how much each device covers in dollars, months, and megawatts. Without that arithmetic, a cost-shifting risk remains.

3. A national pledge is not self-executing. The Ratepayer Protection Pledge is important because it says the right thing at the national level. But the pledge must become filed tariffs, service agreements, collateral, minimum bills, and local fiscal commitments. Otherwise it remains a political promise rather than a settlement mechanism.

4. Good regulation prices risk instead of freezing investment. A rigid 24/7 load should face a higher minimum bill and stronger collateral. A flexible load that can curtail during grid stress, bring reliable resources, or post a strong parent guarantee can deserve lower charges. The public interest is not served by either blank checks or blanket hostility.

5. Local opposition is not always anti-technology, and support is not always naive. Residents may be asking a reasonable balance sheet question: if local government receives development credit and future tax claims, who bears the water, power, road, emergency-service, and electric-bill risks? Supporters can answer that question by pointing to enforceable obligations rather than asking the public to trust the growth story.

25 Conclusion: Clear the Ledger Before Taking Sides

The Money View does not say that AI data centers are good or bad. It says the public debate should begin with promises that can be checked.

Before asking whether a community supports an AI data center, ask:

- Whose balance sheet gets the new asset?
- Whose balance sheet gets the new liability?
- Who provides the liquidity backstop?
- Who bears stranded asset risk?
- How many dollars of risk are covered by minimum bills, collateral, and exit fees?
- Who must pay cash when the bill comes due?
- If the forecast is wrong, where does the loss land?

Only after those questions are answered can the public debate become honest.

> In a polarized debate, slogans pull people apart. A balance sheet graph puts the argument back on one page.

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Change Note

Technical Toolkit v1.0 RC3 adds the release-candidate version seal; the Liquidity Happy Hours / Money View Lab brand split; the public product ladder for civic handout, reporter kit, advocate/PUC toolkit, and legal appendix; the front-door packet; the task-first module directory; the three-tier public-risk trigger; default materiality trigger menu; the Causation Gate, Causation Evidence Matrix, four-stage ledger-line test, cluster/queue ledger, cluster cost allocation rule, worked cluster-allocation example, acceleration-value formula, and screening-versus-adjudicative allocation warning for project-caused exposure; the unit and NPV protocol; layered adjudication rules with distributional consequences; the eight-status Ledger Certificate taxonomy; public-welfare override conditions and the distinction between policy-support and payment-settlement override; required stress-case matrix; evidence-band framing before numeric haircuts; screening-only haircut rules with cap/floor limits; the regional capacity causation module; the PJM do-not-pro-rate warning and mini-use example; the no-cross-ledger-offset and public-welfare dashboard distinction; benefit categories separating settlement, system, and strategic benefits; supporter evidence protocol, the fair-opponent-loss standard, and the fair-supporter-loss standard; the utility forecast-error sharing mechanism; institutional remedy-boundary, authority-prototype, and sample-order-package tables; interim approval options with deadline, reopener, true-up, and carrying-cost rules; source-pinning register with partial citation lockdown; sample order conditions with more cautious legal language; PJM status-check tables; the accreditation boundary and flexibility template; the cross-jurisdiction ledger statement versions; the Distributional Adequacy Rule, materiality triggers, and threshold-source rule; local burden categories, three local governance layers, local nexus rule, and the permit-compliance-is-not-settlement warning; the early abuse-warning box; the depolarization research appendix and minimal pilot design; the fully filled hypothetical worksheet with unit distinctions; and annotated references explaining how key sources are used. It should be treated as the authoritative technical companion for this release candidate.